THE MATHEMATICAL WORKS OF T. J. KACZYNSKI: A UNIFIED ACCOUNT AND FORMALIZATION BLUEPRINT

Abstract. We give a single, self-contained account, in traditional mathematical language, of every result in the following seven sources: the doctoral dissertation *Boundary functions* (1967) [3]; the papers [1] (described only by its review MR0176080), [2], [5], [6], [7]; and the note [4] on a problem of Alan Sutcliffe. The one result known to us only through a review (Theorem 1 of [1], on homeomorphisms of the disk) is given a complete proof by the plane-topological methods of the other papers. Proofs that the originals leave to the reader (the "readily verifiable facts" of [6], the existence of special functions, the explicit Sutcliffe solutions, the extension theorems for Baire functions) are written out. Classical results from the literature that the originals cite are isolated as numbered *Facts*. The document is organised as a blueprint for a Lean 4 formalization: every numbered item names the Lean declarations that formalize it and the items it depends on.

Contents

1. Introduction 2
1.1. The sources 2
1.2. How the document is organised 2
1.3. Conventions 3
2. Sutcliffe's problem 3
3. Borel classes and Baire functions 5
3.1. Classes of sets 5
3.2. Classes of functions 5
3.3. Functions into the sphere 8
3.4. Points of continuity 9
3.5. Higher Borel and Baire classes 9
4. Arcs, crosscuts and transfer 11
4.1. Arcs and curvilinear limits 11
4.2. Crosscuts 12
4.3. Transfer 13
5. Sets of curvilinear convergence 14
The half-plane lemmas of the dissertation 16
6. Boundary functions of continuous functions and homeomorphisms 16
6.1. Homeomorphisms 17
7. Realizing sets of curvilinear convergence and boundary functions 18
7.1. Regions 18
7.2. Special families and special functions 19
7.3. The function g 20
7.4. The function h 21
7.5. The function k 22
7.6. Assembly 23
8. Boundary functions of discontinuous functions 24
8.1. Baire functions 24

Date: October 2026.

8.2. Lebesgue measure 25
9. Bounded harmonic functions 27
10. A cube with a non-Borel set of curvilinear convergence 30
Appendix A. The dissertation's special families 31
Appendix B. Concordance with the sources 32
References 33

1. Introduction

1.1. **The sources.** Seven documents are covered.

- [4] (1968): an elementary number-theoretic note settling a conjecture of Sutcliffe on integers that are multiplied by an integer when their digits are reversed. Section 2.
- [1] (1965), known to us only through its review. Its theorems concern boundary functions of homeomorphisms, continuous, Baire and measurable functions in a disk. Theorems 2–6 of [1] reappear, with proofs, in the dissertation [3] (which says so explicitly); Theorem 1 (on homeomorphisms) does not, and is proved here in Section 6 by the crosscut methods of [2, 3].
- [2] (1966): a short proof of McMillan's theorem that the set of curvilinear convergence of a continuous function is of type $F_{\sigma\delta}$, and a generalization to cluster sets. Section 5.
- [3] (1967): the dissertation. Its Introduction (Baire functions) is in Section 3; Chapter I in Sections 5, 6, 7; Chapter II in Section 8.
- [6] (1969): every $F_{\sigma\delta}$ set is the set of curvilinear convergence of a continuous function with a prescribed boundary function (McMillan's problem); appendix on Baire functions into the sphere. Sections 7 and 3.
- [5] (1969): boundary functions of bounded harmonic functions. Section 9.
- [7] (1969): a bounded continuous function in a cube whose set of curvilinear convergence is not Borel. Section 10.

A concordance between the numbered results of the sources and the items of this document is given in Appendix B.

1.2. **How the document is organised.** Several results occur in more than one source in slightly different settings (the disk $\mathbb{D}$, the upper half-plane $\mathbb{H}$, the strip-like region Q). We prove each result once, in the most convenient model, and transfer it to the others by an explicit homeomorphism (Lemma 4.18). When two sources prove the same statement by different arguments, we give the argument that is shorter or more general, and say so in the concordance.

Facts. Results from the literature that the sources use without proof are stated as numbered *Facts*, with their provenance. In the formalization they are separate Lean theorems, proved or left as `sorry`; they are never axioms. Some of them are deep (the Jordan curve theorem, Kierst's theorem on asymptotic values of holomorphic functions); the dependency map shows exactly which results rest on which Facts.

Departures from the sources. We record here the places where our proof differs in substance from the source.

(i) The step (B)(III) of [6] cites a theorem of Bagemihl and McMillan; we use instead the approximation theorem of the dissertation (Theorem 3.14), as [6] itself suggests.
(ii) The proof of (F)(I) in [6] cites an invariance-of-domain corollary from [19]; we write down the inverse maps explicitly (Lemma 3.18).
(iii) The extension theorems for Baire-one functions (cited from [13] in [3] and used in Appendix (b),(c) of [6]) are proved (Theorem 3.16).

(iv) The "Figure 5" argument of [3] and the monotonicity of first hitting points are derived from a single imported crosscut theorem (Fact 4.10).
(v) Sutcliffe's Theorems 2 and 3, used in [4], are proved with explicit solutions (Propositions 2.3, 2.4).
(vi) In the existence of special functions (statement (XX) of [6]) we give a non-recursive construction: all disjointness constraints turn out to be independent of the other member's size.

1.3. **Conventions.** $\mathbb{N} = \{1, 2, 3, \dots\}$. "Countable" includes "finite". We identify $\mathbb{R}^2$ with $\mathbb{C}$ and write points as $z = x + iy = \langle x, y\rangle$. We use

$$\mathbb{D} = \{z : |z| < 1\}, \qquad C = \{z : |z| = 1\}, \qquad \mathbb{H} = \{z : \operatorname{Im} z > 0\},$$
$$Q = \{\langle x, y\rangle : -1 \le x \le 1,\ 0 < y \le 1\}, \qquad Q^\circ = (-1, 1) \times (0, 1), \qquad X = (-1, 1) \times \{0\}.$$

The real axis is identified with $\mathbb{R}$; in Sections 7 we identify X with $(-1, 1)$. For $n \in \mathbb{N}$, $\mathbb{H}_n = \{0 < y < 1/n\}$, $\overline{\mathbb{H}}_n = \{0 \le y \le 1/n\}$ and $X_n = \{y = 1/n\}$. $\mathbb{S}^2 = \{v \in \mathbb{R}^3 : |v| = 1\}$ is the Riemann sphere with the Euclidean metric. $S(r, p)$ is the open ball of radius r about p; $\operatorname{Cl} E$ or $\overline{E}$ is closure, E^* or $\operatorname{int} E$ interior (for subsets of C or of $\mathbb{R}$, relative to C or $\mathbb{R}$). For an interval or arc I, I^* is its relative interior; an interval or arc is *nondegenerate* if it has more than one point.

2. Sutcliffe's problem

Source: [4], which uses Theorems 2 and 3 of [8]; we prove these too.

Definition 2.1 (digit solutions)**.** For an integer $n \ge 2$ and nonnegative integers $a_h, \dots, a_0$ write $(a_h, \dots, a_1, a_0)_n = a_h n^h + \cdots + a_1 n + a_0$. An $(h+1)$*-digit solution for* n is a collection of integers $n, a_h, \dots, a_0, k$ with

$$n \ge 2, \quad k > 1, \quad a_h > 0, \quad a_0 > 0, \quad 0 \le a_i \le n-1\ (0 \le i \le h), \qquad k\,(a_h, \dots, a_1, a_0)_n = (a_0, a_1, \dots, a_h)_n.$$

We say *there is an* $(h+1)$*-digit solution for* n if such a_i, k exist. Thus a 2-digit solution is (a, c, k) with $k(an + c) = cn + a$, and a 3-digit solution is (a, b, c, k) with $k(an^2 + bn + c) = cn^2 + bn + a$, where $1 \le a, c \le n - 1$, $0 \le b \le n - 1$, $k \ge 2$.

Lemma 2.2 (basic inequalities)**.** *For a 2-digit solution (a, c, k) for n, and for a 3-digit solution (a, b, c, k) for n, we have $k < n$ and $a < c$.*

Proof. In the 3-digit case $an^2+bn+c \ge n^2$ because $a \ge 1$, and $cn^2+bn+a \le (n-1)(n^2+n+1) = n^3-1 < n^3$; so $kn^2 \le k(an^2+bn+c) < n^3$ and $k < n$. Since $k \ge 2$, $cn^2+bn+a \ge 2(an^2+bn+c) > an^2 + bn + c$, i.e. $(c - a)(n^2 - 1) > 0$, so $c > a$. The 2-digit case is the same with $an + c \ge n$, $cn + a \le n^2 - 1$ and $(c - a)(n - 1) > 0$. □

Proposition 2.3 (Sutcliffe, Theorem 2, first half)**.** *If there is a 2-digit solution for n, then $n+1$ is not prime.*

Proof. Let $k(an + c) = cn + a$. Modulo $p = n + 1$ we have $n \equiv -1$, so $k(c - a) \equiv a - c$, i.e. $p \mid (k + 1)(c - a)$. By Lemma 2.2, $0 < c - a < n < p$ and $3 \le k + 1 \le n < p$. If p were prime it would divide one of the two factors, which is impossible. □

Proposition 2.4 (explicit solutions)**.** *Let $n \ge 2$ and suppose $n + 1 = de$ with integers $d \ge 3$ and $e \ge 2$. Put $a = e - 1$, $c = n - e$, $k = d - 1$. Then (a, c, k) is a 2-digit solution for n and $(a, n - 1, c, k)$ is a 3-digit solution for n.*

Proof. Digits: $a = e-1 \geq 1$; $e \leq (n+1)/3 \leq n-1$ gives $c = n-e \geq 1$; clearly $a, c \leq n-1$; and $k = d-1 \geq 2$. Using $n+1 = de$:

$$k(an+c) = (d-1)\big((e-1)n + n - e\big) = (d-1)e(n-1),$$
$$cn + a = (n-e)n + e - 1 = (n-1)(n+1-e) = (n-1)(de-e) = (d-1)e(n-1),$$
$$k\big(an^2 + (n-1)n + c\big) = (d-1)\big((e-1)n^2 + n^2 - n + n - e\big) = (d-1)e(n^2-1),$$
$$cn^2 + (n-1)n + a = (n-e)n^2 + n^2 - n + e - 1 = (n^2-1)(n+1-e) = (d-1)e(n^2-1). \quad \square$$

Lemma 2.5 (factorization)**.** *If $n \geq 4$ and $n+1$ is not prime, then $n+1 = de$ with $d \geq 3$ and $e \geq 2$.*

Proof. Write $n+1 = uv$ with $2 \leq u \leq v$. Then $v \geq 3$, since $v = 2$ would force $n+1 = 4$. Take $d = v$, $e = u$. $\square$

Theorem 2.6 (Sutcliffe, Theorem 2)**.** *Let $n \geq 4$. There is a 2-digit solution for n if and only if $n+1$ is not prime.*

Proof. One direction is Proposition 2.3; the other is Lemma 2.5 with Proposition 2.4. $\square$

Theorem 2.7 (Sutcliffe, Theorem 3)**.** *If there is a 2-digit solution for n, then there is a 3-digit solution for n.*

Proof. By Lemma 2.2, $1 < k < n$, so $n \geq 3$. For $n = 3$ we would have $k = 2$ and $2(3a+c) = 3c+a$, i.e. $c = 5a \geq 5 > n-1$; impossible. So $n \geq 4$, $n+1$ is not prime by Proposition 2.3, and Lemma 2.5 with Proposition 2.4 gives a 3-digit solution. (In $\mathbb{N}$-valued arithmetic the expressions $n(a-1)$, $a-b+c$ below are to be read in $\mathbb{Z}$.) $\square$

Lemma 2.8 ([4, Lemma 1])**.** *There are no natural numbers a, c, n with $n \geq 2$, $a < c \leq n-1$, $a + c \geq n+1$, and $n(a-1)+c \mid n(c-1)+a$.*

Proof. Suppose there are, and write

$$k\big(n(a-1)+c\big) = n(c-1)+a, \qquad k \geq 1. \tag{2.1}$$

(Here $a \geq 2$ because $a \geq n+1-c \geq 2$, so both sides are positive and $k \geq 1$.) Modulo $n-1$ we have $n \equiv 1$, so $k(a-n+c) \equiv c-n+a$, i.e. $n-1 \mid (k-1)(a-n+c)$. Since $a+c \geq n+1$, $a-n+c > 0$. If $k = 1$ then $n(a-1)+c = n(c-1)+a$, so $c-a = n(c-a)$ and $n = 1$, contrary to $n \geq 2$. Hence $k > 1$, $(k-1)(a-n+c) > 0$, and so $(k-1)(a+c-n) \geq n-1$. Since $c-n \leq -1$, $(k-1)(a-1) \geq (k-1)(a+c-n) \geq n-1$. Therefore

$$k\big(n(a-1)+c\big) = nk(a-1)+kc \geq n(k-1)(a-1)+kc \geq n(n-1)+kc \geq nc+kc > n(c-1)+a,$$

using $c \leq n-1$ and $kc > 0 \geq a-n$. This contradicts (2.1). $\square$

Lemma 2.9 ([4, Lemma 2])**.** *If $n = 2$ or $n = 3$, there is no 3-digit solution for n.*

Proof. By Lemma 2.2, $1 < k < n$, which is impossible for $n = 2$. For $n = 3$ we get $k = 2$ and $2(9a+3b+c) = 9c+3b+a$, i.e. $0 = 17a+3b-7c \geq 17+3b-14 > 0$, a contradiction. $\square$

Theorem 2.10 (Kaczynski [4]; Sutcliffe's conjecture)**.** *Let $n \geq 2$. If there is a 3-digit solution for n, then there is a 2-digit solution for n.*

Proof. By Lemma 2.9 we may assume $n > 3$, and by Theorem 2.6 we may assume that $p = n+1$ is prime. Let (a, b, c, k) be a 3-digit solution:

$$k(an^2 + bn + c) = cn^2 + bn + a. \tag{2.2}$$

Modulo p, $n \equiv -1$, so $k(a-b+c) \equiv c-b+a$ and $p \mid (k-1)(a-b+c)$. If $p \nmid a-b+c$ then $p \mid k-1$; since $k-1 > 0$ this gives $k-1 \geq p$, impossible because $k < n$ (Lemma 2.2). Hence

$p \mid a-b+c$. Now $-p < -n < 3-n \le a-b+c \le 2n-2 < 2p$, so $a-b+c = ep$ with $e \in \{0,1\}$. Substituting $b = a+c-ep$,

$$an^2+bn+c = a(n^2+n)+c(n+1)-epn = p\big((a-e)n+c\big), \qquad cn^2+bn+a = p\big((c-e)n+a\big),$$

so (2.2) becomes $k\big((a-e)n+c\big) = (c-e)n+a$. If $e=1$ then $k\big(n(a-1)+c\big) = n(c-1)+a$; moreover $0 \le b = a+c-p$ gives $a+c \ge n+1$, and $a < c \le n-1$ by Lemma 2.2. This is impossible by Lemma 2.8. Hence $e=0$ and $k(an+c) = cn+a$ with $1 \le a, c \le n-1$ and $k > 1$: a 2-digit solution. □

Corollary 2.11 ([4])**.** *If $n > 3$, there is a 3-digit solution for n if and only if $n+1$ is not prime.*

Proof. If $n+1$ is not prime, Theorem 2.6 gives a 2-digit solution and Theorem 2.7 a 3-digit one. Conversely a 3-digit solution gives a 2-digit solution (Theorem 2.10), and then $n+1$ is not prime (Theorem 2.6). □

Remark 2.12. [4] notes that a similar result was obtained independently by Prasert Na Nagara.

3. Borel classes and Baire functions

Sources: [3, Introduction, §3], [6, Appendix]. Throughout, M is a metric space.

3.1. Classes of sets.

Definition 3.1 (F_σ, G_δ, $F_{\sigma\delta}$, ambiguous sets)**.** A subset of a topological space is F_σ if it is a countable union of closed sets, G_δ if it is a countable intersection of open sets, $F_{\sigma\delta}$ if it is a countable intersection of F_σ sets, and *ambiguous* if it is both F_σ and G_δ. For $L \subseteq M$ we write $F_\sigma(L)$, $G_\delta(L)$, $F_{\sigma\delta}(L)$ for the corresponding classes of the subspace L.

Lemma 3.2 (closure properties)**.**
(a) *In M, open sets are F_σ, closed sets are G_δ, and countable sets are F_σ.*
(b) *F_σ is closed under countable unions and finite intersections; G_δ under countable intersections and finite unions; $F_{\sigma\delta}$ under countable intersections and finite unions. The complement of an F_σ set is G_δ and conversely.*
(c) *For $L \subseteq M$: $E \in F_\sigma(L)$ iff $E = P \cap L$ for some F_σ set P of M; likewise for G_δ and $F_{\sigma\delta}$.*
(d) *If L is $F_{\sigma\delta}$ (resp. G_δ) in M and $E \in F_{\sigma\delta}(L)$ (resp. $G_\delta(L)$), then E is $F_{\sigma\delta}$ (resp. G_δ) in M. Every F_σ set is $F_{\sigma\delta}$, and every G_δ set is $F_{\sigma\delta}$.*
(e) *Homeomorphisms preserve all these classes.*

Proof. (a) $U = \bigcup_k \{x : \operatorname{dist}(x, M \setminus U) \ge 1/k\}$ for $U \ne M$ open; complements give the rest, and points are closed. (b) $(\bigcup_n A_n) \cap (\bigcup_m B_m) = \bigcup_{n,m} A_n \cap B_m$, and $(\bigcap_n P_n) \cup (\bigcap_m R_m) = \bigcap_{n,m}(P_n \cup R_m)$; the rest is De Morgan. (c) Closed sets of L are traces of closed sets of M, and traces commute with countable unions and intersections. (d) $E = P \cap L$ with P in the class, and the class is closed under finite intersections; a G_δ set $\bigcap_n U_n$ is $F_{\sigma\delta}$ by (a). (e) is clear. □

3.2. Classes of functions.

Definition 3.3 (function classes)**.** Let L be a metric space and Y a metric space.
(a) $f : L \to Y$ is *of class F_σ* if $f^{-1}(U) \in F_\sigma(L)$ for every open $U \subseteq Y$.
(b) f is *of Baire class ≤ 1*, written $f \in \mathcal{B}_1(L,Y)$, if there are continuous $f_n : L \to Y$ with $f_n \to f$ pointwise on L.
(c) f is *of honorary Baire class ≤ 2* if there are a countable $N \subseteq L$ and $\psi \in \mathcal{B}_1(L,Y)$ with $f = \psi$ on $L \setminus N$.
(d) (Baire classes of all orders.) f is of Baire class 0 if it is continuous; for an ordinal $\xi \ge 1$, f is of Baire class ξ if it is the pointwise limit of a sequence f_n, each of Baire class η_n for some $\eta_n < \xi$. (So class ξ contains all classes $\eta < \xi$ when $\xi \ge 1$.)

Lemma 3.4 (reduction; [3, Lemma 1])**.** *Let E, F be F_σ subsets of M. There are disjoint F_σ sets $A \subseteq E$, $B \subseteq F$ with $A \cup B = E \cup F$. In particular $E \setminus F \subseteq A$ and $F \setminus E \subseteq B$.*

Proof. Write $E = \bigcup_n E_n$, $F = \bigcup_n F_n$ with E_n, F_n closed, and put

$$A = \bigcup_n \Big(E_n \setminus \bigcup_{j<n} F_j \Big), \qquad B = \bigcup_n \Big(F_n \setminus \bigcup_{j \le n} E_j \Big).$$

Each term is closed minus closed, i.e. closed $\cap$ open, hence F_σ (Lemma 3.2). If $x \in (E_n \setminus \bigcup_{j<n} F_j) \cap (F_m \setminus \bigcup_{j\le m} E_j)$ then $m < n$ is impossible because $x \in F_m$, and $m \ge n$ is impossible because $x \in E_n$; so $A \cap B = \emptyset$. If $x \in E \cup F$, let n be least with $x \in E_n \cup F_n$. If $x \in E_n$ then $x \in A$; otherwise $x \in F_n \setminus \bigcup_{j\le n} E_j \subseteq B$. □

Corollary 3.5 (finite reduction)**.** *If $P_1, \dots, P_q$ are F_σ subsets of M, there are pairwise disjoint F_σ sets $R_i \subseteq P_i$ with $\bigcup_i R_i = \bigcup_i P_i$. If $\bigcup_i P_i = M$, each R_i is ambiguous.*

Proof. Induction on q. Reduce P_1 against $P' = P_2 \cup \cdots \cup P_q$ (Lemma 3.4), getting disjoint F_σ sets $A \subseteq P_1$, $B \subseteq P'$ with $A \cup B = P_1 \cup P'$; apply the induction hypothesis to $P_2, \dots, P_q$, getting $R_2', \dots, R_q'$, and put $R_1 = A$, $R_i = R_i' \cap B$. If the R_i cover M, the complement of each is the finite union of the others, hence F_σ. □

Theorem 3.6 (Sierpiński [20])**.** *If $E \subseteq F \subseteq M$ with E a G_δ set and F an F_σ set, there is an ambiguous set K with $E \subseteq K \subseteq F$.*

Proof. $P = M \setminus E$ and F are F_σ with $P \cup F = M$. Reduce them (Lemma 3.4): $A \subseteq P$, $B \subseteq F$ disjoint F_σ with $A \cup B = M$. Put $K = B$. Then $K \subseteq F$, $K = M \setminus A \supseteq M \setminus P = E$, and K is F_σ with F_σ complement A. □

Lemma 3.7. *Every $f \in \mathcal{B}_1(L, Y)$ is of class F_σ.*

Proof. Let $f_n \to f$ with f_n continuous, and $U \subsetneq Y$ open. Put $V_k = \{y : \operatorname{dist}(y, Y \setminus U) \ge 1/k\}$, a closed subset of U. Then $f^{-1}(U) = \bigcup_{k,N} \bigcap_{n \ge N} f_n^{-1}(V_k)$: if $\operatorname{dist}(f(x), Y \setminus U) > 2/k$ then eventually $f_n(x) \in V_k$; conversely if $f_n(x) \in V_k$ for $n \ge N$ then $f(x) \in V_k \subseteq U$. □

Lemma 3.8 (Banach's Hilfssatz 3)**.** *If $f : L \to Y$ is of class F_σ and takes only the values $a_1, \dots, a_q$ (distinct), then the sets $E_j = f^{-1}(a_j)$ are ambiguous in L and partition L.*

Proof. With r less than half the least distance between the a_j, $E_j = f^{-1}(S(r, a_j))$ is F_σ, and its complement $\bigcup_{i \ne j} E_i$ is F_σ too. □

Lemma 3.9. *If $f, g : L \to \mathbb{R}$ are finitely valued and of class F_σ, so are $f + g$ and $f - g$.*

Proof. By Lemma 3.8 the level sets of f and g are ambiguous; every preimage under $f \pm g$ is a finite union of sets $f^{-1}(a) \cap g^{-1}(b)$, which are F_σ. □

Lemma 3.10 ([3, Lemma 2])**.** *Let $L \subseteq M$ and let E be ambiguous in L. There are continuous $f_n : M \to [0, 1]$ with $f_n \to \chi_E$ pointwise on L.*

Proof. Write $E = E_1 \cap L$ and $L \setminus E = F_1 \cap L$ with E_1, F_1 F_σ in M. Reduce: disjoint F_σ sets $A \subseteq E_1$, $B \subseteq F_1$ with $A \cup B = E_1 \cup F_1 \supseteq L$. Then $E = A \cap L$ and $L \setminus E = B \cap L$ (a point of E is not in F_1, hence not in B, hence in A). Write $A = \bigcup A_n$, $B = \bigcup B_n$ with $A_n \subseteq A_{n+1}$, $B_n \subseteq B_{n+1}$ closed, and let f_n be the Urysohn function $\operatorname{dist}(x, B_n)/(\operatorname{dist}(x, A_n) + \operatorname{dist}(x, B_n))$ (if $A_n = \emptyset$ take $f_n = 0$; if $B_n = \emptyset$ take $f_n = 1$), which is 1 on A_n and 0 on B_n. For $x \in L$, eventually $x \in A_n$ or eventually $x \in B_n$. □

Lemma 3.11 ([3, Lemma 3])**.** *If $f : L \to \mathbb{R}$ is of class F_σ and finitely valued, and $L \subseteq M$, there are continuous $f_n : M \to \mathbb{R}$ with $f_n \to f$ pointwise on L.*

Proof. $f = \sum_j a_j \chi_{E_j}$ with E_j ambiguous (Lemma 3.8); take $f_n = \sum_j a_j f_n^{(j)}$ with $f_n^{(j)} \to \chi_{E_j}$ from Lemma 3.10. □

Lemma 3.12 ([3, Lemma 4], Banach's Hilfssatz 4)**.** *If $f : L \to \mathbb{R}$ is bounded and of class F_σ, then for every $k \in \mathbb{N}$ there is a finitely valued $g : L \to \mathbb{R}$ of class F_σ with $|g - f| < 1/k$ and $\sup|g| \le \sup|f| + 1/k$.*

Proof. Cover $[-\sup|f|, \sup|f|]$ by the finitely many intervals $U_j = ((j-1)/k, (j+1)/k)$ that meet it. The sets $f^{-1}(U_j)$ are F_σ in L and cover L; Corollary 3.5 gives a partition of L into ambiguous $R_j \subseteq f^{-1}(U_j)$. Put $g = j/k$ on R_j. Then $|g - f| < 1/k$, and $g^{-1}(W)$ is a finite union of R_j's for every W. □

Lemma 3.13 ([3, Lemma 5])**.** *Let $L \subseteq M$ and let $f : L \to \mathbb{R}$ be of class F_σ. There are continuous $f_j : M \to \mathbb{R}$ with $f_j \to f$ pointwise on L.*

Proof. Bounded f. Choose by Lemma 3.12 finitely valued g_n of class F_σ with $\|g_n - f\|_\infty \le 2^{-n}$; put $h_1 = g_1$, $h_n = g_n - g_{n-1}$. Each h_n is finitely valued of class F_σ, and $\|h_n\| \le 2^{-n+2}$ for $n \ge 2$. By Lemma 3.11 choose continuous $h_{n,j} \to h_n$ on L, and let $k_{n,j}$ be $h_{n,j}$ clamped to $[-\|h_n\|, \|h_n\|]$; still $k_{n,j} \to h_n$ on L. Then $f_j = \sum_n k_{n,j}$ converges uniformly, so it is continuous. For $x \in L$ and $m \in \mathbb{N}$,

$$|f_j(x) - f(x)| \le \sum_{n>m} \|h_n\| + \sum_{n \le m} |k_{n,j}(x) - h_n(x)| + |g_m(x) - f(x)|,$$

which is small for m large and then j large.

General f. $\arctan \circ f$ is bounded of class F_σ; take continuous $G_j \to \arctan \circ f$ on L, clamp G_j into $[-\pi/2 + 1/j, \pi/2 - 1/j]$ to get H_j (still convergent on L, since the target values lie in the open interval), and put $f_j = \tan \circ H_j$. □

Theorem 3.14 ([3, Lemma 6])**.** *Let $L \subseteq M$ and $f : L \to \mathbb{R}^m$. The following are equivalent: (i) $f \in \mathcal{B}_1(L, \mathbb{R}^m)$; (ii) f is of class F_σ; (iii) there are continuous $f_n : M \to \mathbb{R}^m$ with $f_n \to f$ pointwise on L.*

Proof. (i)⇒(ii) is Lemma 3.7; (iii)⇒(i) by restriction. (ii)⇒(iii): each coordinate $f_i = \pi_i \circ f$ is of class F_σ; apply Lemma 3.13 to each. (Conversely the coordinates being of class F_σ implies f is: an open set of $\mathbb{R}^m$ is a countable union of open boxes, and $f^{-1}(\prod_i I_i) = \bigcap_i f_i^{-1}(I_i)$.) □

Lemma 3.15 (uniform limits)**.** *If $S_n : L \to Y$ are of class F_σ and $S_n \to F$ uniformly, then F is of class F_σ.*

Proof. Let $U \subsetneq Y$ be open and $U_k = \{y : \operatorname{dist}(y, Y \setminus U) > 2/k\}$; for each k choose n_k with $\operatorname{dist}(S_{n_k}(x), F(x)) < 1/k$ for all x. Then $F^{-1}(U) = \bigcup_k S_{n_k}^{-1}(U_k)$. □

Theorem 3.16 (extension of Baire-one functions)**.** *Let $f : L \to \mathbb{R}^m$ be of Baire class ≤ 1, $L \subseteq M$.*

(a) If L is G_δ in M and f is bounded, f extends to a bounded $F \in \mathcal{B}_1(M, \mathbb{R}^m)$.
(b) If $M \setminus L$ is countable, f extends to some $F \in \mathcal{B}_1(M, \mathbb{R}^m)$.

Proof. It suffices to treat $m = 1$ (extend coordinatewise).

(a) *Finitely valued f of class F_σ on a G_δ set L.* By Lemma 3.8, $f = \sum_j a_j \chi_{E_j}$ with $E_1, \dots, E_q$ an ambiguous partition of L; write $E_j = P_j \cap L$ with P_j F_σ in M. The sets $P_1, \dots, P_q, M \setminus L$ are F_σ and cover M; let $R_1, \dots, R_q, R_0$ be the disjoint F_σ cover given by Corollary 3.5. Put $F = a_j$ on R_j and $F = 0$ on R_0. For $x \in L \cap E_j$, $x \notin R_0$, and $x \in R_i$ forces $x \in P_i \cap L = E_i$, so $i = j$ and $F(x) = f(x)$. F is finitely valued of class F_σ and $\sup|F| \le \max|a_j|$.

Bounded f. f is of class F_σ (Lemma 3.7). With g_n, h_n as in the proof of Lemma 3.13, extend each h_n to a finitely valued H_n of class F_σ with $\sup|H_n| \le \sup|h_n|$. The partial sums $S_N = \sum_{n\le N} H_n$ are finitely valued of class F_σ and converge uniformly to some F, which is bounded, of class F_σ (Lemma 3.15), hence in $\mathcal{B}_1(M,\mathbb{R})$ (Theorem 3.14 with $L = M$); and $F|_L = \lim g_N = f$.

(b) Let $\tau(v) = v/(1+|v|)$, a homeomorphism of $\mathbb{R}^m$ onto the open unit ball B. Since $M \setminus L$ is countable, L is G_δ; by (a) extend $\tau \circ f$ to a bounded $G \in \mathcal{B}_1(M,\mathbb{R}^m)$. The set $Z = G^{-1}(\mathbb{R}^m \setminus B)$ lies in $M \setminus L$, so it is countable. Put $G' = G$ off Z and $G' = 0$ on Z. For open $U \subseteq B$, $G'^{-1}(U) = G^{-1}(U) \cup (Z \text{ or } \emptyset)$ is F_σ. So $G' : M \to B$ is of class F_σ, hence so is $\tau^{-1} \circ G' : M \to \mathbb{R}^m$, which is therefore in $\mathcal{B}_1$ (Theorem 3.14) and extends f. □

3.3. **Functions into the sphere.** We use the folding device of [6, (F)(I)]. Write points of $\mathbb{S}^2$ as $\langle x,y,z\rangle$ and let $T(z) = 4z^3 - 3z$.

Definition 3.17 (the fold)**.** Let $s = 2^{-1/2}$ and

$$U = \{-s < z \le 1\}, \quad V = \{-1 \le z < s\}, \quad Z_U = \{s < z \le 1\}, \quad Z_V = \{-1 \le z < -s\},$$

$$\Phi\langle x,y,z\rangle = \begin{cases} \langle x(4z^2-1),\, y(4z^2-1),\, T(z)\rangle, & |z| \ge s, \\ \langle x,y,-z\rangle, & |z| \le s, \end{cases}$$

all subsets of $\mathbb{S}^2$. Let $p_1 = \langle 1,0,0\rangle$ and let $\sigma : \mathbb{S}^2 \setminus \{p_1\} \to \mathbb{R}^2$ be stereographic projection from p_1.

Lemma 3.18 (properties of the fold)**.** *(i) Φ is a continuous map $\mathbb{S}^2 \to \mathbb{S}^2$. (ii) Φ restricts to homeomorphisms $Z_U \to U$ and $Z_V \to V$. (iii) Each fibre of Φ has at most three points. (iv) U, V are open and $U \cup V = \mathbb{S}^2$. (v) $Z_U \cup Z_V \subseteq \mathbb{S}^2 \setminus \{p_1\}$ and $\sigma(Z_U \cup Z_V)$ is bounded.*

Proof. On $\mathbb{S}^2$, $(x^2+y^2)(4z^2-1)^2 + T(z)^2 = (1-z^2)(16z^4-8z^2+1) + z^2(16z^4-24z^2+9) = 1$, so Φ maps into $\mathbb{S}^2$. At $z = \pm s$ we have $4z^2 - 1 = 1$ and $T(\pm s) = \mp s$, so the two formulas agree and Φ is continuous by the pasting lemma. $T'(z) = 12z^2 - 3 > 0$ for $|z| > 1/2$, so T is a strictly increasing bijection $[s,1] \to [-s,1]$ and $[-1,-s] \to [-1,s]$. Given $\langle x',y',z'\rangle \in U$, put $z = T^{-1}(z') \in (s,1]$, $c = 4z^2 - 1 \in (1,3]$, $x = x'/c$, $y = y'/c$; the identity above shows $\langle x,y,z\rangle \in \mathbb{S}^2$, and this is a continuous inverse of $\Phi|_{Z_U}$. Likewise for Z_V. Φ is injective on each of Z_U, $\{|z| \le s\}$, Z_V, which cover $\mathbb{S}^2$; this gives (iii). (iv) is clear. (v): $\mathrm{Cl}(Z_U \cup Z_V) = \{|z| \ge s\}$ is compact and misses p_1. □

Lemma 3.19 (fold factorisation)**.** *Let $f : M \to \mathbb{S}^2$ be of class F_σ. There is $\psi : M \to Z_U \cup Z_V$ with $\Phi \circ \psi = f$ such that $\sigma \circ \psi : M \to \mathbb{R}^2$ is bounded and of class F_σ.*

Proof. $f^{-1}(U)$ is F_σ and $M \setminus f^{-1}(V)$ is G_δ, and $M \setminus f^{-1}(V) \subseteq f^{-1}(U)$ since $U \cup V = \mathbb{S}^2$. Theorem 3.6 gives an ambiguous K with $M \setminus f^{-1}(V) \subseteq K \subseteq f^{-1}(U)$; put $L = M \setminus K$, so $f(K) \subseteq U$, $f(L) \subseteq V$. Let $\Phi_U = \Phi|_{Z_U} : Z_U \to U$ and $\Phi_V = \Phi|_{Z_V} : Z_V \to V$ (homeomorphisms by Lemma 3.18), and define $\psi = \Phi_U^{-1} \circ f$ on K, $\psi = \Phi_V^{-1} \circ f$ on L. For $G \subseteq \mathbb{S}^2 \setminus \{p_1\}$ open, $\psi^{-1}(G) = [K \cap f^{-1}(\Phi_U(Z_U \cap G))] \cup [L \cap f^{-1}(\Phi_V(Z_V \cap G))]$, and $\Phi_U(Z_U \cap G)$ is open in U, hence in $\mathbb{S}^2$; so $\psi^{-1}(G)$ is F_σ. Boundedness is Lemma 3.18(v). □

Theorem 3.20 ([6, Appendix, Theorem (a)]; cf. [3, Theorem 1])**.** *A function $f : M \to \mathbb{S}^2$ is in $\mathcal{B}_1(M,\mathbb{S}^2)$ if and only if it is of class F_σ.*

Proof. One direction is Lemma 3.7. Conversely, take ψ from Lemma 3.19; by Theorem 3.14 there are continuous $\psi_n : M \to \mathbb{R}^2$ with $\psi_n \to \sigma \circ \psi$. Then $\Phi \circ \sigma^{-1} \circ \psi_n$ are continuous and converge to $\Phi \circ \psi = f$. □

Theorem 3.21 ([6, Appendix, Theorem (c)]). *If E is a G_δ subset of M and $f \in \mathcal{B}_1(E, \mathbb{S}^2)$, then f extends to some $F \in \mathcal{B}_1(M, \mathbb{S}^2)$.*

Proof. By Lemmas 3.7 and 3.19, applied in the metric space E, write $f = \Phi \circ \psi$ with $\sigma \circ \psi : E \to \mathbb{R}^2$ bounded of class F_σ, hence in $\mathcal{B}_1$. By Theorem 3.16(a) it extends to a bounded $G \in \mathcal{B}_1(M, \mathbb{R}^2)$; let continuous $G_n \to G$ on M. Then $\Phi \circ \sigma^{-1} \circ G_n$ are continuous and converge on M to $F = \Phi \circ \sigma^{-1} \circ G$, and $F|_E = \Phi \circ \psi = f$. □

Theorem 3.22 (honorary Baire class 2; [3, Theorem 2], [6, Appendix, Theorem (b)]). *Let Y be $\mathbb{R}^m$ or $\mathbb{S}^2$ and $f : M \to Y$. The following are equivalent.*

(1) f is of honorary Baire class ≤ 2.
(2) There is a countable $N \subseteq M$ such that $f|_{M \setminus N}$ is of class F_σ.
(3) There is a countable $N \subseteq M$ such that $f^{-1}(F) \setminus N$ is G_δ in M for every closed $F \subseteq Y$.

Proof. (1)⇒(2): restrict the Baire-one function and use Lemma 3.7. (2)⇒(1): $f|_{M \setminus N} \in \mathcal{B}_1$ by Theorem 3.14 or 3.20, and $M \setminus N$ is G_δ; extend it to M by Theorem 3.16(b) or 3.21. (2)⇔(3): $f^{-1}(F) \setminus N = (M \setminus N) \setminus (f|_{M \setminus N})^{-1}(Y \setminus F)$; a subset of the G_δ set $M \setminus N$ is G_δ in M iff it is G_δ in $M \setminus N$, and complements in $M \setminus N$ exchange $F_\sigma(M \setminus N)$ and $G_\delta(M \setminus N)$. □

Lemma 3.23 (invariance of honorary Baire class 2). *If $f : M \to Y$ is of honorary Baire class ≤ 2, so are $f|_{M'}$ for $M' \subseteq M$ and $f \circ h$ for a homeomorphism $h : M'' \to M$.*

Proof. Restrict (resp. compose with h) the Baire-one function, its continuous approximants and the countable set. □

3.4. **Points of continuity.**

Theorem 3.24 (Baire, Kuratowski [16, p. 301]). *If A and Y are metric spaces and $f \in \mathcal{B}_1(A, Y)$, then the set of points of A at which f is discontinuous is of the first category in A.*

Proof. Let $f_n \to f$ be continuous. For $\varepsilon > 0$ and $N \in \mathbb{N}$ the set $F_N^\varepsilon = \{x : \operatorname{dist}(f_n(x), f_m(x)) \leq \varepsilon \ \forall n, m \geq N\}$ is closed and $\bigcup_N F_N^\varepsilon = A$. Put $G^\varepsilon = \bigcup_N \operatorname{int} F_N^\varepsilon$. Then $A \setminus G^\varepsilon \subseteq \bigcup_N (F_N^\varepsilon \setminus \operatorname{int} F_N^\varepsilon)$, a countable union of closed nowhere dense sets. If $x \in \operatorname{int} F_N^\varepsilon$, then $\operatorname{dist}(f, f_N) \leq \varepsilon$ on a neighbourhood of x, and f_N is continuous, so $\operatorname{dist}(f(x'), f(x)) \leq 3\varepsilon$ for x' near x and the oscillation of f at x is at most 6ε. Hence the discontinuity set is contained in $\bigcup_k (A \setminus G^{1/k})$. □

Corollary 3.25. *If P is a nonempty complete metric space, Y a metric space and $f \in \mathcal{B}_1(P, Y)$, then f is continuous at some point of P.*

Proof. By the Baire category theorem P is not of the first category in itself. □

Lemma 3.26. *A subset of $A \subseteq M$ that is of the first category in A is of the first category in M.*

Proof. If $N \subseteq A$ is nowhere dense in A and an open $O \neq \emptyset$ of M lies in $\operatorname{Cl}_M N$, then O meets A and $O \cap A \subseteq \operatorname{Cl}_A N$, a contradiction. □

3.5. **Higher Borel and Baire classes.** These are needed only in Section 8.

Definition 3.27 ([3, §3]). $P^1(M)$ is the family of open subsets of M and $Q^1(M)$ that of closed subsets. For an ordinal $\xi > 1$, $P^\xi(M)$ consists of the countable unions of members of $\bigcup_{\eta<\xi} Q^\eta(M)$, and $Q^\xi(M)$ of the countable intersections of members of $\bigcup_{\eta<\xi} P^\eta(M)$. A function $f : M \to Y$ is *of class* P^ξ if $f^{-1}(U) \in P^\xi(M)$ for every open $U \subseteq Y$.

Lemma 3.28. *(a) $E \in Q^\xi$ iff $M \setminus E \in P^\xi$. (b) P^ξ and Q^ξ are closed under finite unions and intersections, P^ξ under countable unions, Q^ξ under countable intersections. (c) If $\eta < \xi$ then $P^\eta \cup Q^\eta \subseteq P^\xi \cap Q^\xi$. (d) $P^2(M)$ is the class of F_σ sets and $Q^2(M)$ that of G_δ sets. (e) Continuous preimages preserve each class. (f) For $L \subseteq M$, $P^\xi(L) = \{D \cap L : D \in P^\xi(M)\}$. (g) Every member of every class is a Borel set.*

Proof. Transfinite induction on ξ. (c) for $\eta = 1$ uses that in a metric space open sets are F_σ and closed sets are G_δ (Lemma 3.2(a)); for $\eta > 1$ it is immediate from the definitions. For (b), $(\bigcup A_n) \cap (\bigcup B_m) = \bigcup A_n \cap B_m$, and $A_n \in Q^{\eta_1}$, $B_m \in Q^{\eta_2}$ both lie in $Q^{\max(\eta_1,\eta_2)}$ by (c). The rest is routine. □

Lemma 3.29. *If $f : M \to Y$ is of Baire class ξ and $g : K \to M$ is continuous, then $f \circ g$ is of Baire class ξ. In particular restrictions to subspaces keep the class. If $1 \le \xi \le \xi'$, Baire class ξ is contained in Baire class ξ'.*

Proof. Induction on ξ; for the last claim use constant sequences. □

Lemma 3.30. *If $f : M \to Y$ is of Baire class ξ (Y any metric space), then f is of class $P^{\xi+1}$. In particular it is Borel measurable.*

Proof. For $\xi = 0$ preimages of open sets are open. For $\xi \ge 1$ let $f_n \to f$ with f_n of class $\eta_n < \xi$. With V_k as in Lemma 3.7, $f^{-1}(U) = \bigcup_{k,N} \bigcap_{n \ge N} f_n^{-1}(V_k)$; by induction $f_n^{-1}(V_k) \in Q^{\eta_n+1} \subseteq Q^\xi$, so the intersections lie in Q^ξ and the union in $P^{\xi+1}$. □

Fact 3.31 (Banach [12])**.** *Let Y be a separable, arcwise connected metric space and $\xi \ge 2$. If $f : M \to Y$ is of class $P^{\xi+1}$, then f is of Baire class ξ.*

Fact 3.32 (Lebesgue–Hausdorff–Banach)**.** *Let Y be a separable, arcwise connected metric space. Every Borel-measurable $f : M \to Y$ is of Baire class ξ for some countable ordinal ξ.*

Lemma 3.33 ([3, Lemma 18])**.** *Write $A \approx B$ if $A \setminus B$ and $B \setminus A$ are countable. This is an equivalence relation; $A \approx E$ implies $S \setminus A \approx S \setminus E$; and if $A_n \approx E_n$ for n in a countable set, then $\bigcap A_n \approx \bigcap E_n$ and $\bigcup A_n \approx \bigcup E_n$.*

Proof. $(\bigcap A_n) \setminus (\bigcap E_n) \subseteq \bigcup_n (A_n \setminus E_n)$, and similarly for the rest. □

Lemma 3.34 ([3, Lemma 19], [2, Lemma 1])**.** *If $\mathcal{I}$ is a family of nondegenerate intervals of $\mathbb{R}$ (resp. nondegenerate closed arcs of C), then $\bigcup_{I \in \mathcal{I}} I \setminus \bigcup_{I \in \mathcal{I}} I^*$ is countable. In particular every union of nondegenerate intervals is $\approx$ an open set.*

Proof. $\bigcup I^*$ is open, so it is the disjoint union of its countably many components J_n. If x_0 lies in $\bigcup I$ but not in $\bigcup I^*$, then x_0 is an endpoint of some I_0; $I_0^* \subseteq J_n$ for some n, so $x_0 \in \overline{J_n} \setminus J_n$ is an endpoint of J_n. There are countably many such endpoints. □

Lemma 3.35 (two-sided limit points)**.** *For $F \subseteq \mathbb{R}$, the set of points of $\overline{F}$ that are not two-sided limit points of F is countable. (Here x is a* two-sided limit point *of F if every interval $(x-r, x)$ and $(x, x+r)$, $r > 0$, meets F.)*

Proof. If $x \in \overline{F}$ and $(x, x+r) \cap F = \emptyset$, then $(x, x+r)$ misses $\overline{F}$, so x is the left endpoint of a component of $\mathbb{R} \setminus \overline{F}$. There are countably many components; argue symmetrically on the left. □

Lemma 3.36 ([3, Lemma 20])**.** *Let h be weakly increasing on a nonempty $E \subseteq \mathbb{R}$ with $|x - h(x)| \le 1$ on E. Then h extends to a weakly increasing $h_1 : \mathbb{R} \to \mathbb{R}$.*

Proof. Let $e = \inf E$. For $x > e$ put $h_1(x) = \sup h((-\infty, x] \cap E)$, which is finite since $h(t) \le t + 1 \le x + 1$ there. If $e > -\infty$ put $h_1(x) = \inf h(E)$ for $x \le e$, finite since $h \ge e - 1$ on E. Monotonicity and $h_1|_E = h$ are immediate. □

Lemma 3.37 ([3, Lemma 21])**.** *If $f:\mathbb{R}\to Y$ is of Baire class ξ and $h:\mathbb{R}\to\mathbb{R}$ is weakly increasing, there is a countable $N\subseteq\mathbb{R}$ such that $(f\circ h)|_{\mathbb{R}\setminus N}$ is of Baire class ξ.*

Proof. Let N be the (countable) set of discontinuities of h and apply Lemma 3.29 to $h|_{\mathbb{R}\setminus N}$. □

Lemma 3.38 ([3, Lemma 22])**.** *Let Y be a separable arcwise connected metric space, E a metric space, $\xi\geq 2$, and $\varphi:E\to Y$ such that for every open $U\subseteq Y$ there is $T\in P^{\xi+1}(E)$ with $\varphi^{-1}(U)\subseteq T\subseteq\varphi^{-1}(\overline{U})$. Then φ is of Baire class ξ.*

Proof. Let $\mathcal{B}$ be a countable base of Y. For open W let $\mathcal{B}(W)=\{B\in\mathcal{B}:\overline{B}\subseteq W\}$; then $W=\bigcup_{\mathcal{B}(W)}B=\bigcup_{\mathcal{B}(W)}\overline{B}$ (for $p\in W$ with $S(r,p)\subseteq W$ choose $p\in B\subseteq S(r/2,p)$). With $T(B)$ as in the hypothesis, $\varphi^{-1}(W)=\bigcup_{B\in\mathcal{B}(W)}T(B)\in P^{\xi+1}(E)$. Conclude by Fact 3.31. □

4. Arcs, crosscuts and transfer

4.1. Arcs and curvilinear limits.

Definition 4.1 (arcs)**.** A *path* in $\mathbb{R}^n$ is a continuous $g:[0,1]\to\mathbb{R}^n$; a *simple arc* is the image of an injective path, and its *endpoints* are $g(0),g(1)$. Let $\Omega\subseteq\mathbb{R}^n$ be open and $p\in\partial\Omega$. An *arc at p* (in Ω) is a simple arc γ with one endpoint p and $\gamma\setminus\{p\}\subseteq\Omega$. A *crosscut* of $\mathbb{D}$ is a simple arc whose endpoints are distinct points of C and whose other points lie in $\mathbb{D}$; a crosscut of $\mathbb{H}$ is defined in the same way with $\mathbb{R}$ in place of C.

Definition 4.2 (curvilinear convergence, boundary functions, cluster sets)**.** Let Ω be open, $B\subseteq\partial\Omega$, Y a metric space and $f:\Omega\to Y$. For an arc γ at p, $f(z)\to q$ *along* γ means $f(z)\to q$ as $z\to p$ with $z\in\gamma\setminus\{p\}$. The *set of curvilinear convergence* of f (relative to B) is

$$\mathrm{CC}(f)=\{p\in B:\ \text{there are an arc }\gamma\text{ at }p\text{ and }q\in Y\text{ with }f\to q\text{ along }\gamma\}.$$

If $E\subseteq B$, a function $\varphi:E\to Y$ is a *boundary function* for f if for every $p\in E$ there is an arc γ at p with $f\to\varphi(p)$ along γ. The *cluster set* of f along γ is $C(f,\gamma)=\{q:\exists z_n\in\gamma\setminus\{p\},\ z_n\to p,\ f(z_n)\to q\}$. The pairs (Ω,B) used are $(\mathbb{D},C)$, $(\mathbb{H},\mathbb{R})$ and (Q°,X), and in Section 10 the open cube with its whole boundary.

Fact 4.3 (Tietze [22])**.** *If g is a path in $\mathbb{R}^n$ with $g(0)\neq g(1)$, then $g([0,1])$ contains a simple arc with endpoints $g(0)$ and $g(1)$.*

Corollary 4.4 (paths may replace arcs)**.** *Let Ω be open, $p\in\partial\Omega$, and $g:[0,1]\to\mathbb{R}^n$ continuous with $g(1)=p$ and $g([0,1))\subseteq\Omega$. Then $g([0,1])$ contains an arc γ at p, and for every $f:\Omega\to Y$: if $f(g(t))\to q$ as $t\to 1$ then $f\to q$ along γ; and $C(f,\gamma)$ is contained in the set of limits of $f(g(t_n))$ with $t_n\to 1$.*

Proof. Fact 4.3 gives a simple arc $\gamma\subseteq g([0,1])$ from $g(0)$ to p; since $g(t)\neq p$ for $t<1$, $\gamma\setminus\{p\}\subseteq\Omega$. If $z_k\in\gamma\setminus\{p\}$ and $z_k\to p$, write $z_k=g(t_k)$ with $t_k<1$; a subsequence of t_k bounded away from 1 would have a limit $t^*<1$ with $g(t^*)=p$, which is impossible. So $t_k\to 1$. □

Lemma 4.5 (subarcs and exits)**.** *(a) If γ is an arc at p and W is a neighbourhood of p, there is an arc $\gamma'\subseteq\gamma\cap W$ at p; limits and cluster sets along γ' and γ coincide. If $f\to q$ along γ and $V\ni q$ is open, some such γ' has $f(\gamma'\setminus\{p\})\subseteq V$. (b) If g is a path, $w=g(t_0)$, and $|g(t_1)-w|\geq r>0$, there is a closed interval J with endpoints t_0 and some t' between t_0 and t_1 such that $g(J)\subseteq\overline{S(r,w)}$ and $|g(t')-w|=r$; in particular $\operatorname{diam}g(J)\geq r$. (c) If γ is an arc at p in $\mathbb{H}$ and n is large enough that γ meets X_n, then the subarc of γ from p to its first point on X_n is an arc at p contained in $\overline{\mathbb{H}}_n$ with all points other than its endpoints in $\mathbb{H}_n$.*

Proof. Parametrize and take initial segments; for (b) let t' be the first parameter between t_0 and t_1 with $|g(t')-w|\geq r$, and use continuity. □

Lemma 4.6 (cluster sets)**.** *Let K be a compact metric space, $f : \Omega \to K$ continuous and γ an arc at p. Then $C(f,\gamma) = \bigcap_{\gamma'} \overline{f(\gamma' \setminus \{p\})}$, the intersection over the subarcs γ' of γ at p; it is nonempty, compact and connected; and $f \to q$ along γ iff $C(f,\gamma) = \{q\}$. If $\Theta : K \to K'$ is continuous into a metric space, then $C(\Theta \circ f, \gamma) = \Theta(C(f,\gamma))$.*

Proof. The sets $\overline{f(\gamma' \setminus \{p\})}$ are compact, connected (images of half-open arcs) and decrease as γ' shrinks; a decreasing intersection of nonempty compact connected sets is nonempty, compact and connected. The claim about limits is the usual compactness argument. For the last claim: $\supseteq$ by continuity of Θ; if $\Theta(f(z_n)) \to q'$ with $z_n \to p$ along γ, a subsequence of $f(z_n)$ converges in the compact K to some $q \in C(f,\gamma)$, and $\Theta(q) = q'$. □

Fact 4.7 (Tietze extension)**.** *A bounded continuous function from a closed subset of a metric space to $\mathbb{C}$ extends to a bounded continuous function on the whole space.*

Fact 4.8 (hyperspace)**.** *If K is a compact metric space, the space $\mathcal{C}(K)$ of nonempty closed subsets of K with the Hausdorff metric is compact.*

Fact 4.9 (Baire category theorem)**.** *A complete metric space is not of the first category in itself; in it, a countable intersection of dense open sets is dense.*

4.2. **Crosscuts.**

Fact 4.10 (crosscut theorem)**.** *Let σ be a crosscut of $\mathbb{D}$ with endpoints $a \neq b$, and let A_1, A_2 be the two closed arcs of C with endpoints a, b. Then $\mathbb{D} \setminus \sigma$ has exactly two components U_1, U_2, and the frontier of U_i is $\sigma \cup A_i$ $(i = 1, 2)$. In particular $\overline{U_i} \cap C = A_i$ and $\sigma \subseteq \overline{U_1} \cap \overline{U_2}$.*

Lemma 4.11 (disjoint components have disjoint boundary arcs)**.** *Let σ, τ be crosscuts of $\mathbb{D}$, U a component of $\mathbb{D} \setminus \sigma$ and V a component of $\mathbb{D} \setminus \tau$, with $U \cap V = \emptyset$. Then the arcs $\overline{U} \cap C$ and $\overline{V} \cap C$ have disjoint relative interiors.*

Proof. Let p be interior to both arcs. Then p is not an endpoint of σ or τ, so $p \notin \sigma \cup \tau$, and there is $r > 0$ with $S(r,p) \cap (\sigma \cup \tau) = \emptyset$. The convex set $S(r,p) \cap \mathbb{D}$ lies in one component of $\mathbb{D} \setminus \sigma$; as $p \in \overline{U}$ it meets U, so it lies in U. Likewise it lies in V, contradicting $U \cap V = \emptyset$. □

Definition 4.12 (charts)**.** (a) The Cayley map $\kappa(z) = (z - i)/(z + i)$.
(b) $\Psi_{\mathbb{H}}\langle x, y\rangle = \tan(\pi x/2) + i\, y/(1 - y)$ for $|x| < 1$, $0 \leq y < 1$.
(c) $\Psi_{\mathbb{D}}\langle x, y\rangle = 1 - 1/w$ with $w = \frac{1}{2}(1 + y) + i \tan(\pi x/2)$, for $|x| < 1$, $0 \leq y < 1$.
(d) $\mathcal{E}_h(z) = e^{\pi z/h}$ for $h > 0$.

Lemma 4.13 (charts)**.** *(a) κ is a homeomorphism of $\mathbb{H} \cup \mathbb{R}$ onto $(\mathbb{D} \cup C) \setminus \{1\}$ mapping $\mathbb{H}$ onto $\mathbb{D}$ and $\mathbb{R}$ onto $C \setminus \{1\}$; $|\kappa(z) - 1| \to 0$ iff $|z| \to \infty$. (b) $\Psi_{\mathbb{H}}$ is a homeomorphism of $Q^\circ \cup X$ onto $\mathbb{H} \cup \mathbb{R}$ mapping X onto $\mathbb{R}$ increasingly. (c) $\Psi_{\mathbb{D}}$ is a homeomorphism of $Q^\circ \cup X$ onto $G^\circ \cup (C \setminus \{1\})$, where $G^\circ = \{z \in \mathbb{D} : \operatorname{Re}(1 - z)^{-1} < 1\}$, mapping X onto $C \setminus \{1\}$. The set $G'' = \{z \in \mathbb{D} : \operatorname{Re}(1 - z)^{-1} \leq 3/4\}$ is closed in $\mathbb{D}$, contains a neighbourhood in $\mathbb{D}$ of each point of $C \setminus \{1\}$, and is disjoint from the radius $[0, 1)$. (d) $\mathcal{E}_h$ maps $\{0 \leq \operatorname{Im} z \leq h\}$ homeomorphically onto $(\mathbb{H} \cup \mathbb{R}) \setminus \{0\}$, the line $\operatorname{Im} z = 0$ increasingly onto $(0, \infty)$ and the line $\operatorname{Im} z = h$ decreasingly onto $(-\infty, 0)$.*

Proof. (a) $|\kappa(z)| < 1$ iff $|z - i| < |z + i|$ iff $\operatorname{Im} z > 0$; the inverse is $w \mapsto i(1 + w)/(1 - w)$. (b) is a product of homeomorphisms $(-1, 1) \to \mathbb{R}$ and $[0, 1) \to [0, \infty)$. (c) $z \mapsto (1 - z)^{-1}$ maps $\mathbb{D}$ onto $\{\operatorname{Re} w > \frac{1}{2}\}$ and $C \setminus \{1\}$ onto $\{\operatorname{Re} w = \frac{1}{2}\}$ (since $|1 - 1/w| < 1 \iff |w - 1| < |w|$); $\Psi_{\mathbb{D}}$ is the inverse of this map composed with a homeomorphism of $Q^\circ \cup X$ onto $\{\frac{1}{2} \leq \operatorname{Re} w < 1\}$. Near a point $x \in C \setminus \{1\}$, $\operatorname{Re}(1 - z)^{-1}$ is close to $\frac{1}{2}$; on $[0, 1)$ it is ≥ 1. (d) is elementary. □

Lemma 4.14 (crosscuts of the half-plane)**.** *Let σ be a crosscut of $\mathbb{H}$ with endpoints $a < b$. Then $\mathbb{H} \setminus \sigma$ has exactly two components, a bounded one V with $\overline{V} \cap \mathbb{R} = [a, b]$, and an unbounded one U with $\overline{U} \cap \mathbb{R} = \mathbb{R} \setminus (a, b)$; moreover $\sigma \subseteq \overline{U} \cap \overline{V}$ and $U \supseteq \{\operatorname{Im} z > \max_\sigma \operatorname{Im}\}$.*

Proof. $\kappa(\sigma)$ is a crosscut of $\mathbb{D}$ with endpoints in $C \setminus \{1\}$ (Lemma 4.13(a)); apply Fact 4.10. The component whose closed arc avoids 1 has closure a compact subset of $\overline{\mathbb{D}} \setminus \{1\}$, so its preimage V is bounded, and κ maps $[a, b]$ onto that arc. The half-plane above σ is connected, unbounded and disjoint from σ, so it lies in U. □

Lemma 4.15 (order of disjoint arcs in a strip)**.** *Let $h > 0$, $S = \{0 < \operatorname{Im} z < h\}$, and let α, α' be disjoint simple arcs, α from $x \in \mathbb{R}$ to p with $\operatorname{Im} p = h$, α' from $x' \in \mathbb{R}$ to p' with $\operatorname{Im} p' = h$, all other points in S. If $x < x'$ then $\operatorname{Re} p < \operatorname{Re} p'$.*

Proof. By Lemma 4.13(d), $\mathcal{E}_h(\alpha)$ is a crosscut of $\mathbb{H}$ from $u = e^{\pi x/h}$ to $v = -e^{\pi \operatorname{Re} p/h}$, and $\mathcal{E}_h(\alpha')$ one from $u' > u$ to $v' = -e^{\pi \operatorname{Re} p'/h} \neq v$. By Lemma 4.14 the bounded component V of $\mathbb{H} \setminus \mathcal{E}_h(\alpha)$ has $\overline{V} \cap \mathbb{R} = [v, u]$. The connected set $\mathcal{E}_h(\alpha') \setminus \{u', v'\}$ lies in one component; its closure contains $u' \notin [v, u]$, so it lies in the unbounded component U, and then $v' \in \overline{U} \cap \mathbb{R} = \mathbb{R} \setminus (v, u)$. As $v' < 0 < u$ and $v' \neq v$, $v' < v$, i.e. $\operatorname{Re} p' > \operatorname{Re} p$. □

Lemma 4.16 (geometry near a boundary point)**.** *Let $x \in C$ and $0 < \delta < 2$. (a) $\mathbb{D} \cap S(\delta, x)$ is convex. (b) $\mathbb{D} \setminus \overline{S(\delta, x)}$ is connected and has $-x$ in its closure. (c) If $x' \in C$ and $|x' - x| < \delta$, the shorter closed arc of C from x to x' lies in $S(\delta, x)$. Any arc of C between w_1, w_2 has length at least $|w_1 - w_2|$. (d) If $A_1, \dots, A_m$ are closed arcs of C with pairwise disjoint relative interiors, their lengths add up to at most 2π. (The length of an arc is the Lebesgue measure of the corresponding set of angles in $[0, 2\pi)$.)*

Proof. (a) Intersection of convex sets. (b) For $z \in \mathbb{D}$ the convex function $t \mapsto |(1-t)z - tx - x|^2$ has derivative $2(1 - |z|^2) > 0$ at $t = 0$, hence increases on $[0, 1]$; so the segment $[z, -x)$ stays in $\mathbb{D} \setminus \overline{S(\delta, x)}$ and ends in the convex set $\mathbb{D} \cap S(2 - \delta, -x)$, which lies in $\mathbb{D} \setminus \overline{S(\delta, x)}$. (c) On an arc of angle at most π the chord from an endpoint increases with the angle; chords are shorter than arcs. (d) The interiors are disjoint measurable sets of angles and the endpoints are null. □

Lemma 4.17. *An open subset of $\mathbb{R}^n$ has countably many components; they are open and path-connected, and a connected subset of the open set lies in one of them.*

Proof. $\mathbb{R}^n$ is locally path-connected and second countable. □

4.3. **Transfer.**

Lemma 4.18 (transfer)**.** *Let Ω, Ω' be open, $B \subseteq \partial\Omega$, $B' \subseteq \partial\Omega'$, and let $\Psi : \Omega \cup B \to \Omega' \cup B'$ be a homeomorphism with $\Psi(\Omega) = \Omega'$. For $f : \Omega' \to Y$: γ is an arc at $p \in B$ iff $\Psi(\gamma)$ is an arc at $\Psi(p)$; $f \circ \Psi \to q$ along γ iff $f \to q$ along $\Psi(\gamma)$, and the cluster sets agree; hence $\mathrm{CC}(f \circ \Psi) = \Psi^{-1}(\mathrm{CC}(f))$, and φ is a boundary function for f on E' iff $\varphi \circ \Psi$ is one for $f \circ \Psi$ on $\Psi^{-1}(E')$.*

Proof. Arcs at p lie in $\Omega \cup \{p\}$, where Ψ is a homeomorphism. □

Lemma 4.19 (locality)**.** *Let $p \in \partial\Omega \cap \partial\Omega'$ and suppose $W \cap \Omega = W \cap \Omega'$ for an open $W \ni p$. If $f = f'$ on $W \cap \Omega$, then f has a limit q along some arc at p in Ω iff f' has the limit q along some arc at p in Ω'.*

Proof. Lemma 4.5(a). □

5. Sets of curvilinear convergence

Sources: [2]; [3, §4, Lemmas 7–8, Theorem 3]. We work in $\mathbb{D}$ and transfer to $\mathbb{H}$.

Definition 5.1 (accessible points and crosscut arcs)**.** For a crosscut γ of $\mathbb{D}$ with $0 \notin \gamma$, let $L(\gamma) = \overline{V} \cap C$ where V is the component of $\mathbb{D} \setminus \gamma$ not containing 0; by Fact 4.10 it is a nondegenerate closed arc whose endpoints are those of γ. For a domain $\Omega \subseteq \mathbb{D} \setminus \{0\}$ let $\Gamma(\Omega)$ be the set of crosscuts γ with $\gamma \cap \mathbb{D} \subseteq \Omega$,

$$I(\Omega) = \bigcup_{\gamma \in \Gamma(\Omega)} L(\gamma), \qquad I_0(\Omega) = \bigcup_{\gamma \in \Gamma(\Omega)} L(\gamma)^*,$$

and let $\operatorname{acc}(\Omega)$ be the set of $x \in C$ for which there is an arc at x in Ω.

Lemma 5.2 ([2, Lemma 2])**.** *For every domain $\Omega \subseteq \mathbb{D} \setminus \{0\}$, $\operatorname{acc}(\Omega) \setminus I_0(\Omega)$ is countable.*

Proof. $I(\Omega) \setminus I_0(\Omega)$ is countable by Lemma 3.34, so it suffices to show $\operatorname{acc}(\Omega) \subseteq I(\Omega)$ when $\operatorname{acc}(\Omega)$ has two or more points. Let $a \neq a'$ be in $\operatorname{acc}(\Omega)$, with arcs γ, γ' at a, a' in Ω, and join their other endpoints by an arc $\gamma'' \subseteq \Omega$. By Fact 4.3 the union contains a simple arc δ' from a to a'; it is a crosscut with $\delta' \cap \mathbb{D} \subseteq \Omega$ (so $0 \notin \delta'$), and $a \in L(\delta')$. □

Lemma 5.3 ([2, Lemma 3])**.** *Let $\Omega_1, \Omega_2 \subseteq \mathbb{D} \setminus \{0\}$ be domains. If the sets $I_0(\Omega_1) \cap \overline{\operatorname{acc}(\Omega_1)}$ and $I_0(\Omega_2) \cap \overline{\operatorname{acc}(\Omega_2)}$ have a common point, then $\Omega_1 \cap \Omega_2 \neq \emptyset$.*

Proof. Suppose $\Omega_1 \cap \Omega_2 = \emptyset$ and a is a common point. Choose $\gamma_i \in \Gamma(\Omega_i)$ with $a \in L(\gamma_i)^*$, and let U_i, V_i be the components of $\mathbb{D} \setminus \gamma_i$, with $0 \in U_i$. Then $\gamma_1 \cap \mathbb{D}$ and $\gamma_2 \cap \mathbb{D}$ are disjoint, and $\overline{U_i} \cap C = C \setminus L(\gamma_i)^*$.

Suppose $\gamma_1 \cap \mathbb{D} \subseteq V_2$ and $\gamma_2 \cap \mathbb{D} \subseteq V_1$. As $\gamma_1 \cap \mathbb{D} \subseteq \overline{U_1}$ (Fact 4.10) and V_2 is open, U_1 meets V_2; also $0 \in U_1 \cap U_2$. The connected set U_1 meets both components of $\mathbb{D} \setminus \gamma_2$, so it meets $\gamma_2 \cap \mathbb{D} \subseteq V_1$, which is absurd. The connected sets $\gamma_i \cap \mathbb{D}$ each lie in a component of the other complement, so (by symmetry) we may assume $\gamma_2 \cap \mathbb{D} \subseteq U_1$.

Since $a \in \overline{\operatorname{acc}(\Omega_2)}$ and $L(\gamma_1)^*$ is a neighbourhood of a in C, there is $b \in L(\gamma_1)^* \cap \operatorname{acc}(\Omega_2)$. Joining an arc at b in Ω_2 to a point of $\gamma_2 \cap \mathbb{D}$ inside Ω_2 and applying Fact 4.3, we get a simple arc γ from b to a point of $\gamma_2 \cap \mathbb{D}$ with $\gamma \setminus \{b\} \subseteq \Omega_2$. Then $\gamma \setminus \{b\}$ is connected, misses γ_1, and meets U_1; so it lies in U_1 and $b \in \overline{U_1} \cap C = C \setminus L(\gamma_1)^*$, a contradiction. □

Theorem 5.4 (McMillan [18, Theorem 5]; [2, Theorem 1])**.** *Let K be a complete separable metric space and $f : \mathbb{D} \to K$ continuous. Then $\operatorname{CC}(f)$ is an $F_{\sigma\delta}$ subset of C.*

Proof. Let $\{p_k\}$ be dense in K and $\{\theta_m\}$ an enumeration of $\mathbb{Q}$. Put $Q(n,m) = \{re^{it} : 1 - 1/n < r < 1,\ \theta_m < t < \theta_m + 2\pi/n\}$, and let $U(n,m,k,l)$, $l \in \mathbb{N}$, list (with repetitions, or empty) the components of $f^{-1}(S(2^{-n}, p_k)) \cap Q(n,m)$ (Lemma 4.17); put $A(n,m,k,l) = \operatorname{acc} U(n,m,k,l)$ and

$$Y = \bigcap_n \bigcup_{m,k,l} \Big(I_0(U(n,m,k,l)) \cap \overline{A(n,m,k,l)} \Big).$$

$I_0(\cdot)$ is open in C, so Y is $F_{\sigma\delta}$.

$Y \subseteq \operatorname{CC}(f)$. Let $y \in Y$ and choose $U_n = U(n, m[n], k[n], l[n])$ with $y \in I_0(U_n) \cap \overline{\operatorname{acc} U_n}$. By Lemma 5.3, $U_n \cap U_{n+1}$ contains a point z_n; join z_n to z_{n+1} by a path $\gamma_n \subseteq U_{n+1}$. Since $y \in \overline{U_{n+1}} \subseteq \overline{Q(n+1, m[n+1])}$, every point of γ_n lies within $(2\pi+1)/(n+1)$ of y. So the paths γ_n, traversed successively and followed by y, form a path g with $g(1) = y$ and $g([0,1)) \subseteq \mathbb{D}$. As $S(2^{-n}, p_{k[n]})$ and $S(2^{-n-1}, p_{k[n+1]})$ meet, $\rho(p_{k[n]}, p_{k[n+1]}) < 2^{-n+1}$, so $p_{k[n]} \to p$ for some $p \in K$; and $f(\gamma_n) \subseteq S(2^{-n-1}, p_{k[n+1]})$, so $f(g(t)) \to p$ as $t \to 1$. By Corollary 4.4, $y \in \operatorname{CC}(f)$.

$\operatorname{CC}(f) \subseteq \bigcap_n \bigcup_{m,k,l} A(n,m,k,l)$. Let $f \to p'$ along an arc γ_0 at x. Given n, pick k with $\rho(p', p_k) < 2^{-n-1}$ and m with x interior to $\overline{Q(n,m)} \cap C$. By Lemma 4.5 a subarc γ_0' at x

has $\gamma_0' \setminus \{x\} \subseteq Q(n,m) \cap f^{-1}(S(2^{-n}, p_k))$; being connected, it lies in one component, so $x \in A(n,m,k,l)$ for some l.

$\mathrm{CC}(f) \setminus Y$ *is countable.* If $x \in \mathrm{CC}(f) \setminus Y$, there is n with $x \in A(n,m,k,l)$ for some m,k,l but $x \notin I_0(U(n,m,k,l)) \cap \overline{A(n,m,k,l)}$, so $x \in A(n,m,k,l) \setminus I_0(U(n,m,k,l))$, which is countable by Lemma 5.2.

So $\mathrm{CC}(f)$ is the union of an $F_{\sigma\delta}$ set and a countable set, hence $F_{\sigma\delta}$. □

Theorem 5.5 ([2, Theorem 2])**.** *Let K be a compact metric space, $\mathcal{C}(K)$ the space of nonempty closed subsets of K with the Hausdorff metric $\bar\rho$, $\mathcal{E} \subseteq \mathcal{C}(K)$ closed, and $f : \mathbb{D} \to K$ continuous. Then*

$$X_{\mathcal{E}} = \{x \in C : \text{there are an arc } \gamma \text{ at } x \text{ and } E \in \mathcal{E} \text{ with } C(f,\gamma) \subseteq E\}$$

is an $F_{\sigma\delta}$ set.

Proof. If $\mathcal{E} = \emptyset$ there is nothing to prove. $\mathcal{E}$ is compact (Fact 4.8), hence has a dense sequence $P(k)$. For $\varepsilon > 0$ let $\mathcal{S}(\varepsilon, E) = \{a : \mathrm{dist}(a,E) < \varepsilon\}$, which is open, and note that $\bar\rho(E,F) < \varepsilon$ implies $F \subseteq \mathcal{S}(\varepsilon, E)$. Define $Q(n,m)$ as before, $U(n,m,k,l)$ as the components of $f^{-1}(\mathcal{S}(1/n, P(k))) \cap Q(n,m)$, $A(n,m,k,l) = \operatorname{acc} U(n,m,k,l)$, and Y by the same formula; Y is $F_{\sigma\delta}$.

$Y \subseteq X_{\mathcal{E}}$. With U_n as before, compactness gives $P \in \mathcal{E}$ and $n_1 < n_2 < \cdots$ with $P(k[n_j]) \to P$. Lemma 5.3 gives $z_j \in U_{n_j} \cap U_{n_{j+1}}$, and paths $\gamma_j \subseteq U_{n_{j+1}}$ from z_j to z_{j+1}, within $(2\pi+1)/n_{j+1}$ of y; they form a path g ending at y. Let $f(w_s) \to p$ with $w_s \to y$ along the path. Each γ_j is a compact subset of $\mathbb{D}$, so $w_s \in \gamma_{i(s)}$ with $i(s) \to \infty$. Given $\varepsilon > 0$, for s large $f(w_s) \in \mathcal{S}(1/n_{i+1}, P(k[n_{i+1}]))$ with $1/n_{i+1} < \varepsilon/3$ and $\bar\rho(P(k[n_{i+1}]), P) < \varepsilon/3$, and $\rho(f(w_s), p) < \varepsilon/3$; so $\mathrm{dist}(p, P) < \varepsilon$. As P is closed, $p \in P$. By Corollary 4.4 there is an arc γ at y with $C(f,\gamma) \subseteq P$.

$X_{\mathcal{E}} \subseteq \bigcap_n \bigcup_{m,k,l} A(n,m,k,l)$. Let $C(f,\gamma_0) \subseteq P_0 \in \mathcal{E}$. Given n, choose k with $\bar\rho(P_0, P(k)) < 1/n$, so $C(f,\gamma_0) \subseteq \mathcal{S}(1/n, P(k)) =: O$. Some subarc of γ_0 at x is mapped into O: otherwise there are $z_t' \in \gamma_0 \cap S(1/t, x) \cap \mathbb{D}$ with $f(z_t')$ in the compact set $K \setminus O$, and a limit point of $f(z_t')$ would lie in $C(f,\gamma_0) \setminus O$. Conclude as in Theorem 5.4; the countability of $X_{\mathcal{E}} \setminus Y$ is proved in the same way. □

Corollary 5.6 ([2, Corollaries 1–4])**.** *Let $f : \mathbb{D} \to \mathbb{S}^2$ be continuous. Each of the following sets is $F_{\sigma\delta}$:*

1. *(McMillan) $\{x : \exists \gamma \text{ at } x,\ \exists p \in E,\ f \to p \text{ along } \gamma\}$, for closed $E \subseteq \mathbb{S}^2$;*
2. *$\{x : \exists \gamma \text{ at } x,\ \mathrm{diam}\, C(f,\gamma) \le d\}$, for $d \ge 0$;*
3. *$\{x : \exists \gamma \text{ at } x,\ C(f,\gamma) \subseteq E\}$, for closed $E \subseteq \mathbb{S}^2$;*
4. *$\{x : \exists \gamma \text{ at } x,\ C(f,\gamma)$ is an arc of a great circle$\}$, where (since $C(f,\gamma)$ is closed and connected) an "arc" may be a point, a closed arc, or a whole great circle.*

Proof. Apply Theorem 5.5 with $\mathcal{E}$ equal to: (1) $\{\{p\} : p \in E\}$, an isometric copy of E, using that $C(f,\gamma) \subseteq \{p\}$ iff $f \to p$ along γ (Lemma 4.6); (2) $\{F : \mathrm{diam}\, F \le d\}$, closed because $|\mathrm{diam}\, F - \mathrm{diam}\, F'| \le 2\bar\rho(F,F')$, and using $E = C(f,\gamma)$ for the converse; (3) $\{E\}$ (or note the set is empty if $E = \emptyset$); (4) the set of great circles, the continuous image of the compact set of unit normals, together with the connectedness of $C(f,\gamma)$ (Lemma 4.6). In (4) we read "arc" in this broad sense, which is what the connectedness argument indicated in [2] gives. □

Theorem 5.7 (McMillan in the half-plane; [3, Theorem 3])**.** *If K is a complete separable metric space and $f : \mathbb{H} \to K$ is continuous, then $\mathrm{CC}(f) \subseteq \mathbb{R}$ is $F_{\sigma\delta}$.*

Proof. By Lemma 4.18 with $\Psi = \kappa$ (Lemma 4.13(a)), $\mathrm{CC}(f) = \kappa^{-1}(\mathrm{CC}(f \circ \kappa^{-1}) \setminus \{1\})$, and the right side is $F_{\sigma\delta}$ by Theorem 5.4 and Lemma 3.2. □

The half-plane lemmas of the dissertation. For a domain $\Omega \subseteq \mathbb{H}$ and a crosscut γ of $\mathbb{H}$ with endpoints $c < d$ let $L(\gamma) = [c, d]$, the trace on $\mathbb{R}$ of the closure of the bounded component of $\mathbb{H} \setminus \gamma$; let $I(\Omega) = \bigcup L(\gamma)^*$ over crosscuts with $\gamma \cap \mathbb{H} \subseteq \Omega$, and $\operatorname{acc}(\Omega)$ the points of $\mathbb{R}$ accessible by arcs in Ω. These lemmas are the half-plane forms of Lemmas 5.2 and 5.3; they are not used elsewhere.

Lemma 5.8 ([3, Lemma 7])**.** *If $\operatorname{acc}(\Omega) \neq \emptyset$, $a = \inf \operatorname{acc}(\Omega)$ and $b = \sup \operatorname{acc}(\Omega)$, then $I(\Omega) = (a, b)$.*

Proof. If $x \in L(\gamma)^* = (c, d)$ then $c, d \in \operatorname{acc}(\Omega)$, so $a \leq c < x < d \leq b$. Conversely, for $a < x < b$ pick $c', d' \in \operatorname{acc}(\Omega)$ with $c' < x < d'$, and join them by a crosscut through Ω as in Lemma 5.2. □

Lemma 5.9 ([3, Lemma 8])**.** *If $\Omega_1, \Omega_2 \subseteq \mathbb{H}$ are domains and $I(\Omega_1) \cap \overline{\operatorname{acc}(\Omega_1)}$ and $I(\Omega_2) \cap \overline{\operatorname{acc}(\Omega_2)}$ meet, then $\Omega_1 \cap \Omega_2 \neq \emptyset$.*

Proof. As in Lemma 5.3, with U_i the unbounded component of $\mathbb{H} \setminus \gamma_i$ (any two unbounded components meet, since each contains a half-plane $\{\operatorname{Im} z > t\}$) and Lemma 4.14 in place of Fact 4.10. □

6. Boundary functions of continuous functions and homeomorphisms

Sources: [3, §5, Lemma 15, Theorem 5]; [1, Theorems 1–3]; [2, remark after Corollary 4].

Lemma 6.1 ([3, Lemma 15])**.** *Let E be a metric space, Y a separable metric space and $\varphi : E \to Y$. Suppose that for every open $U \subseteq Y$ there are $L \in F_\sigma(E)$ and a countable $N \subseteq E$ with $\varphi^{-1}(U) \subseteq L \subseteq \varphi^{-1}(\overline{U}) \cup N$. Then there is a countable $M \subseteq E$ such that $\varphi|_{E \setminus M}$ is of class F_σ.*

Proof. Let $\mathcal{B}$ be a countable base, choose $L(B), N(B)$ for $B \in \mathcal{B}$, put $M = \bigcup_B N(B)$, $E_0 = E \setminus M$, $\varphi_0 = \varphi|_{E_0}$. For open W let $\mathcal{B}(W) = \{B : \overline{B} \subseteq W\}$, so $W = \bigcup_{\mathcal{B}(W)} B$ (proof of Lemma 3.38). Then

$$\varphi_0^{-1}(W) \subseteq E_0 \cap \bigcup_{\mathcal{B}(W)} L(B) \subseteq E_0 \cap \bigcup_{\mathcal{B}(W)} \left(\varphi^{-1}(\overline{B}) \cup N(B)\right) = E_0 \cap \bigcup_{\mathcal{B}(W)} \varphi^{-1}(\overline{B}) \subseteq \varphi_0^{-1}(W),$$

so $\varphi_0^{-1}(W) = E_0 \cap \bigcup_{\mathcal{B}(W)} L(B) \in F_\sigma(E_0)$. □

Theorem 6.2 ([3, Theorem 5])**.** *Let Y be a separable metric space, $f : \mathbb{H} \to Y$ continuous, $E \subseteq \mathbb{R}$, and $\varphi : E \to Y$ a boundary function for f. There is a countable $M \subseteq E$ such that $\varphi|_{E \setminus M}$ is of class F_σ.*

Proof. By Lemma 6.1 it suffices, given open $U \subseteq Y$, to find $L \in F_\sigma(E)$ and countable N with $\varphi^{-1}(U) \subseteq L \subseteq \varphi^{-1}(\overline{U}) \cup N$. Let $W = Y \setminus \overline{U}$ and

$$E_n = \{x \in \mathbb{R} : \text{some arc } \gamma \text{ at } x \text{ has its other endpoint on } X_n \text{ and } \gamma \setminus \{x\} \subseteq f^{-1}(U)\},$$
$$K = \{x \in \mathbb{R} : \text{some arc } \gamma \text{ at } x \text{ has } \gamma \setminus \{x\} \subseteq f^{-1}(W)\}.$$

If $x \in \varphi^{-1}(U)$, a subarc of the arc at x along which $f \to \varphi(x)$ is mapped into U (Lemma 4.5); cutting it at its first point on a line X_n below its other endpoint shows $x \in E_n$. So $\varphi^{-1}(U) \subseteq \bigcup_n E_n$, and likewise $\varphi^{-1}(W) \subseteq K$.

$K \cap \overline{E_n}$ is countable. Fix n. For $x \in K$ choose (Lemma 4.5) an arc γ_x at x with $\gamma_x \setminus \{x\} \subseteq \mathbb{H}_n \cap f^{-1}(W)$, and let U_x be the component of the open set $\mathbb{H}_n \cap f^{-1}(W)$ containing $\gamma_x \setminus \{x\}$. Let T be the set of points of $K \cap \overline{E_n}$ that are two-sided limit points of E_n. We claim $x \mapsto U_x$ is injective on T, so T is countable (Lemma 4.17). Let $x < y$ in T with $U_x = U_y$. Joining γ_x and γ_y by a path in U_x and applying Fact 4.3, we get a crosscut α' of $\mathbb{H}$ from x to y with $\alpha' \setminus \{x, y\} \subseteq U_x \subseteq \mathbb{H}_n \cap f^{-1}(W)$. Let A be the unbounded component of $\mathbb{H} \setminus \alpha'$ (Lemma 4.14);

since $\max_{\alpha'} \mathrm{Im} < 1/n$, $X_n \subseteq A$. Choose $w \in (x,y) \cap E_n$ and an arc β at w ending on X_n with $\beta \setminus \{w\} \subseteq f^{-1}(U)$. Then $\beta \setminus \{w\}$ is connected, misses α' (as $f^{-1}(U) \cap f^{-1}(W) = \emptyset$) and meets $X_n \subseteq A$, so it lies in A; hence $w \in \overline{A} \cap \mathbb{R} = \mathbb{R} \setminus (x,y)$, a contradiction. The points of $\overline{E_n}$ that are not two-sided limit points of E_n form a countable set (Lemma 3.35). Hence $K \cap \overline{E_n}$ is countable.

Put $N = E \cap K \cap \bigcup_n \overline{E_n}$ (countable) and $L = E \cap \bigcup_n \overline{E_n} \in F_\sigma(E)$. Then $\varphi^{-1}(U) \subseteq L \subseteq N \cup (E \setminus K) \subseteq N \cup (E \setminus \varphi^{-1}(W)) = N \cup \varphi^{-1}(\overline{U})$. □

Corollary 6.3 ([3, Theorem 5, Corollary]; [1, Theorems 2, 3])**.** *Let Y be $\mathbb{R}^m$ or $\mathbb{S}^2$. If $f : \mathbb{H} \to Y$ (resp. $f : \mathbb{D} \to Y$) is continuous, $E \subseteq \mathbb{R}$ (resp. $E \subseteq C$) and $\varphi : E \to Y$ is a boundary function for f, then φ is of honorary Baire class ≤ 2. In particular ([1, Theorems 2, 3]) a finite-valued boundary function on C of a continuous real function in $\mathbb{D}$ is of honorary Baire class $2(C, \mathbb{R})$, and a boundary function on C of a continuous map $\mathbb{D} \to \mathbb{S}^2$ is of honorary Baire class $2(C, \mathbb{S}^2)$.*

Proof. For $\mathbb{H}$ combine Theorem 6.2 with Theorem 3.22(2)⇒(1). For $\mathbb{D}$, Lemma 4.18 with κ shows that $\varphi \circ \kappa$ is a boundary function for $f \circ \kappa$ on $\kappa^{-1}(E \setminus \{1\})$; so $\varphi|_{E \setminus (\kappa(M) \cup \{1\})}$ is of class F_σ for a countable M, and Theorem 3.22 applies. □

Fact 6.4 (Bagemihl's ambiguous point theorem [10])**.** *For any function $f : \mathbb{D} \to \mathbb{S}^2$, the set of $x \in C$ at which there are two arcs along which f has different limits is countable.*

Proposition 6.5 ([2, remark after Corollary 4])**.** *In Corollary 5.6(1) it suffices that E be a G_δ subset of $\mathbb{S}^2$.*

Proof. $A = \mathrm{CC}(f)$ is $F_{\sigma\delta}$ (Theorem 5.4). Choose any boundary function $\varphi : A \to \mathbb{S}^2$. By Corollary 6.3, $\varphi = \psi$ off a countable N_0, with $\psi \in \mathcal{B}_1(A, \mathbb{S}^2)$. Off the countable set Z of Fact 6.4, every arc limit of f at $x \in A$ equals $\varphi(x)$; so the set in question differs from $\psi^{-1}(E)$ by a countable set. Writing $E = \bigcap_j O_j$ with O_j open, $\psi^{-1}(E) = \bigcap_j \psi^{-1}(O_j)$ is $F_{\sigma\delta}$ in A, hence in C; adding or removing countably many points keeps a set $F_{\sigma\delta}$. □

6.1. **Homeomorphisms.** The following is Theorem 1 of [1], of which we have only the statement.

Lemma 6.6. *Let f be a homeomorphism of $\mathbb{D}$ onto $\mathbb{D}$, regarded as a map $\mathbb{D} \to \mathbb{C}$. If $f \to q \in \mathbb{C}$ along an arc at $x \in C$, then $q \in C$.*

Proof. If $q \in \mathbb{D}$ then $z = f^{-1}(f(z)) \to f^{-1}(q) \in \mathbb{D}$ as $z \to x$ along the arc, which is absurd. □

Theorem 6.7 ([1, Theorem 1])**.** *Let f be a homeomorphism of $\mathbb{D}$ onto $\mathbb{D}$, regarded as a map $\mathbb{D} \to \mathbb{C}$, let $E \subseteq C$, and let $\varphi : E \to \mathbb{C}$ be a boundary function for f (so $\varphi(E) \subseteq C$ by Lemma 6.6). Then there is a countable $N \subseteq E$ such that $\varphi|_{E \setminus N}$ is continuous. More precisely, for every $\varepsilon > 0$ the set*

$$S_\varepsilon = \{x \in E : \text{ for every } \delta > 0 \text{ there is } x' \in E \text{ with } 0 < |x' - x| < \delta,\ |\varphi(x') - \varphi(x)| \geq \varepsilon\}$$

has at most $2\pi/\varepsilon$ points.

Proof. Since φ is continuous (relative to E) at every point of $E \setminus \bigcup_k S_{1/k}$, the first claim follows from the second. Let $x_1, \ldots, x_m$ be distinct points of S_ε. Choose $\delta \in (0,1)$ so small that the balls $S(\delta, x_i)$ are pairwise disjoint, and for each i choose $x_i' \in E$ with $0 < |x_i' - x_i| < \delta/2$ and $|\varphi(x_i') - \varphi(x_i)| \geq \varepsilon$. Fix i, and let γ, γ' be arcs at x_i, x_i' along which f tends to $\varphi(x_i), \varphi(x_i')$.

A crosscut near x_i. By Lemma 4.5 choose subarcs $\beta \subseteq \gamma \cap S(\delta/2, x_i)$ at x_i and $\beta' \subseteq \gamma' \cap S(\delta/2, x_i')$ at x_i', with other endpoints $u, v \in \mathbb{D}$. The segment $[u,v]$ lies in the convex set $\mathbb{D} \cap S(\delta, x_i)$ (Lemma 4.16(a)). By Fact 4.3, $\beta \cup [u,v] \cup \beta'$ contains a simple arc σ_i from x_i to x_i', which is a crosscut of $\mathbb{D}$ lying in a closed ball $\overline{S(\delta', x_i)}$ with $\delta' < \delta$. Near x_i, σ_i consists of points

of β (the compact sets $[u,v]$ and β' miss x_i), so $f \to \varphi(x_i)$ along σ_i at x_i; likewise $f \to \varphi(x_i')$ at x_i'. Since $\varphi(x_i) \neq \varphi(x_i')$ lie on C (Lemma 6.6), $\tau_i = f(\sigma_i \cap \mathbb{D}) \cup \{\varphi(x_i), \varphi(x_i')\}$ is a crosscut of $\mathbb{D}$.

The small component. Let R_i be the component of $\mathbb{D} \setminus \sigma_i$ with $\overline{R_i} \cap C$ equal to the shorter arc J_i from x_i to x_i' (Fact 4.10); $J_i \subseteq S(\delta, x_i)$ by Lemma 4.16(c). The connected set $\mathbb{D} \setminus \overline{S(\delta', x_i)}$ (Lemma 4.16(b)) misses σ_i and has $-x_i \notin J_i$ in its closure, so it lies in the other component; hence $R_i \subseteq S(\delta, x_i)$. So $R_1, \dots, R_m$ are pairwise disjoint.

Images. f maps $\mathbb{D} \setminus \sigma_i$ homeomorphically onto $\mathbb{D} \setminus \tau_i$, so $f(R_i)$ is a component of $\mathbb{D} \setminus \tau_i$, and by Fact 4.10 $\overline{f(R_i)} \cap C$ is a closed arc A_i with endpoints $\varphi(x_i), \varphi(x_i')$, of length at least $|\varphi(x_i) - \varphi(x_i')| \geq \varepsilon$ (Lemma 4.16(c)). The sets $f(R_i)$ are pairwise disjoint, so the arcs A_i have pairwise disjoint interiors (Lemma 4.11). Hence $m\varepsilon \leq 2\pi$ by Lemma 4.16(d). □

7. Realizing sets of curvilinear convergence and boundary functions

Sources: [6, (A)–(F)]; [3, §§4–5, Lemmas 9–17, Theorems 4, 6, 7]. In this section $X = (-1,1)$ (identified with $(-1,1) \times \{0\}$), $Q = [-1,1] \times (0,1]$, $Q^\circ = (-1,1) \times (0,1)$, and CC refers to arcs in Q° unless said otherwise. Points of $\mathbb{H}$ are $\langle x, y\rangle$, $y > 0$.

7.1. **Regions.**

Definition 7.1 (Stolz angles, trapezoids, $B(K,\varepsilon,\alpha,\beta)$)**.** For $a \in \mathbb{R}$, $\varepsilon > 0$, $0 < \theta < \pi/2$: $s(a,\varepsilon,\theta) = \{\langle x,y\rangle : 0 < y < \varepsilon,\ |x-a| < y\cot\theta\}$. For a nonempty interval J with endpoints $a \leq b$: $\mathrm{Trap}(J,\varepsilon,\theta) = \{0 < y < \varepsilon,\ a + y\cot\theta < x < b - y\cot\theta\}$ and $\mathrm{Tri}(J,\theta) = \{y \geq 0,\ a + y\cot\theta \leq x \leq b - y\cot\theta\}$. For a nonempty compact $K \subseteq \mathbb{R}$ let $J(K) = [\min K, \max K]$ and let $\mathcal{J}(K)$ be the set of components of $J(K) \setminus K$; for $0 < \beta < \alpha < \pi/2$ put

$$B(K,\varepsilon,\alpha,\beta) = \mathrm{Trap}(J(K),\varepsilon,\alpha) \setminus \bigcup_{I \in \mathcal{J}(K)} \mathrm{Tri}(I,\beta).$$

Lemma 7.2 (description of B)**.** *$\langle x,y\rangle \in B(K,\varepsilon,\alpha,\beta)$ iff $0 < y < \varepsilon$, $\min K + y\cot\alpha < x < \max K - y\cot\alpha$, and $\mathrm{dist}(x,K) < y\cot\beta$.*

Proof. Let $y > 0$ and $x \in J(K)$. If $x \in K$, x lies in no $\mathrm{Tri}(I,\beta)$ (these lie over $\overline{I}$, and at an endpoint a of I we would need $a + y\cot\beta \leq a$). If $x \in I = (a,b) \in \mathcal{J}(K)$, then $x \notin \mathrm{Tri}(I,\beta)$ iff $\min(x-a, b-x) < y\cot\beta$, i.e. iff $\mathrm{dist}(x,K) < y\cot\beta$, and x lies over no other component. □

Lemma 7.3 ([6, (VI)–(XVIII)]; [3, (6)–(17)])**.** *Let K, K_1, K_2 be nonempty compact subsets of $\mathbb{R}$, $0 < \beta < \alpha < \pi/2$, $0 < \theta < \pi/2$. Write $c_\theta = \cot\theta$.*

- *(R1) $s(a,\varepsilon,\theta)$ and $B(K,\varepsilon,\alpha,\beta)$ are open subsets of $\mathbb{H}$.*
- *(R2) $\mathrm{Cl}\, s(a,\varepsilon,\theta) \cap \mathbb{R} = \{a\}$ and $\mathrm{Cl}\, B(K,\varepsilon,\alpha,\beta) \cap \mathbb{R} \subseteq K$.*
- *(R3) If $\varepsilon < \varepsilon'$ and $\theta' < \theta$ then $\mathrm{Cl}\, s(a,\varepsilon,\theta) \cap \mathbb{H} \subseteq s(a,\varepsilon',\theta')$.*
- *(R4) If $a \neq b$ and $\eta \leq |a-b|/(2c_\theta)$, then $s(a,\varepsilon,\theta) \cap s(b,\eta,\theta) = \emptyset$ for all ε.*
- *(R5) If K_1, K_2 split (Definition 7.4), $B(K_1,\varepsilon_1,\alpha,\beta) \cap B(K_2,\varepsilon_2,\alpha,\beta) = \emptyset$.*
- *(R6) If $K_1 \cap K_2 = \emptyset$ and $\eta \leq \mathrm{dist}(K_1,K_2)/(2c_\beta)$, then $B(K_1,\varepsilon,\alpha,\beta) \cap B(K_2,\eta,\alpha,\beta) = \emptyset$ for all ε.*
- *(R7) If $K_1 \subseteq K$, $\varepsilon > \varepsilon_1 > 0$ and $0 < \beta < \beta_1 < \alpha_1 < \alpha < \pi/2$, then $\mathrm{Cl}\, B(K_1,\varepsilon_1,\alpha_1,\beta_1) \cap \mathbb{H} \subseteq B(K,\varepsilon,\alpha,\beta)$.*
- *(R8) If $\alpha < \theta$ and $a \notin J(K)^*$, then $B(K,\varepsilon,\alpha,\beta) \cap s(a,\varepsilon_1,\theta) = \emptyset$.*
- *(R9) If $a \notin K$ and $\eta \leq \mathrm{dist}(a,K)/(c_\theta + c_\beta)$, then $s(a,\eta,\theta) \cap B(K,\varepsilon,\alpha,\beta) = \emptyset$ and $s(a,\varepsilon,\theta) \cap B(K,\eta,\alpha,\beta) = \emptyset$ for all ε.*
- *(R10) If $a \in K \cap J(K)^*$, $\beta < \theta$, and $0 < \eta < \min\{\varepsilon, (a - \min K)/(c_\theta + c_\alpha), (\max K - a)/(c_\theta + c_\alpha)\}$, then $\mathrm{Cl}\, s(a,\eta,\theta) \cap \mathbb{H} \subseteq B(K,\varepsilon,\alpha,\beta)$.*

Proof. All follow from Lemma 7.2 and the triangle inequality. (R1) Strict inequalities between continuous functions. (R2) If $\langle x_n, y_n\rangle \to \langle x, 0\rangle$ with $\operatorname{dist}(x_n, K) < y_n c_\beta$ then $\operatorname{dist}(x, K) = 0$. (R3) On the closure, $y \le \varepsilon < \varepsilon'$ and $|x - a| \le y c_\theta < y c_{\theta'}$. (R4) A common point has $|a - b| < 2y c_\theta < 2\eta c_\theta$. (R5) With splitting point p and $K_1 \le p \le K_2$, points of the first region have $x < \max K_1 \le p$ and those of the second $x > \min K_2 \ge p$. (R6) A common point has $y < \eta$ and $\operatorname{dist}(K_1, K_2) \le \operatorname{dist}(x, K_1) + \operatorname{dist}(x, K_2) < 2y c_\beta$. (R7) On the closure $y \le \varepsilon_1 < \varepsilon$, $\min K + y c_\alpha < \min K_1 + y c_{\alpha_1} \le x$ (similarly on the right), and $\operatorname{dist}(x, K) \le \operatorname{dist}(x, K_1) \le y c_{\beta_1} < y c_\beta$. (R8) If $a \le \min K$, points of the Stolz angle have $x < a + y c_\theta < \min K + y c_\alpha$; symmetrically if $a \ge \max K$. (R9) A common point has $\operatorname{dist}(a, K) \le |x - a| + \operatorname{dist}(x, K) < y(c_\theta + c_\beta)$ with $y < \eta$. (R10) On the closure, $0 < y \le \eta < \varepsilon$, $|x - a| \le y c_\theta$, so $\operatorname{dist}(x, K) \le y c_\theta < y c_\beta$ and $x \ge a - y c_\theta > \min K + y c_\alpha$. □

7.2. **Special families and special functions.**

Definition 7.4 (splitting, special families)**.** Two sets $E_1, E_2 \subseteq \mathbb{R}$ *split* if there is p with $E_1 \le p \le E_2$ or $E_2 \le p \le E_1$ (pointwise). A *special family* is a nonempty countable family $\mathcal{F}$ of compact subsets of X such that any two members are equal, disjoint, or split. Let $\mathcal{F}^2$ be the members with at least two points and $E(\mathcal{F})$ the set of endpoints of the intervals $J(F)$, $F \in \mathcal{F}$, $F \neq \emptyset$.

Two distinct members of a special family meet in at most one point, and such a common point is an endpoint of both hulls. Endpoints of $J(F)$ belong to F.

Lemma 7.5 ([6, (III)]; [3, Lemma 9])**.** *Every F_σ subset E of X is the union of a special family.*

Proof. Write $E = \bigcup_n A_n$ with A_n closed in X, $A_1 = \emptyset$, $A_n \subseteq A_{n+1}$. Every open interval $I \subseteq X$ is the union of countably many compact intervals any two of which split: take a strictly increasing bi-infinite sequence $(c_k)_{k\in\mathbb{Z}}$ in I tending to the endpoints and the intervals $[c_k, c_{k+1}]$. As $X \setminus A_n$ is a disjoint union of countably many open intervals, $X \setminus A_n = \bigcup_j I_{n,j}$ with compact intervals $I_{n,j}$ any two of which split. Let $\mathcal{F} = \{A_1\} \cup \{I_{n,j} \cap A_{n+1}\}$. Its members are compact, and $\bigcup \mathcal{F} = \bigcup_n (A_{n+1} \setminus A_n) = E$. For distinct members $I_{n,j} \cap A_{n+1}$, $I_{n',j'} \cap A_{n'+1}$: if $n < n'$ then $n + 1 \le n'$ and the second lies in $X \setminus A_{n'} \subseteq X \setminus A_{n+1}$, so they are disjoint; if $n = n'$ then $j \neq j'$ and they split because $I_{n,j}, I_{n,j'}$ do. □

Lemma 7.6 ([6, (IV)])**.** *If $A \subseteq X$ is $F_{\sigma\delta}$, there are special families $\mathcal{F}_n$ with $A = \bigcap_n \bigcup \mathcal{F}_n$ such that every member of $\mathcal{F}_{n+1}$ is contained in a member of $\mathcal{F}_n$.*

Proof. Write $A = \bigcap_n A_n$ with A_n F_σ and decreasing, choose special $\mathcal{S}_n$ with $A_n = \bigcup \mathcal{S}_n$, and put $\mathcal{F}_1 = \mathcal{S}_1$, $\mathcal{F}_{n+1} = \{E \cap F : E \in \mathcal{F}_n, F \in \mathcal{S}_{n+1}\}$. For two members $E \cap F$, $E' \cap F'$: if $E \neq E'$ compare E, E', and if $E = E'$ compare F, F'; subsets of disjoint (splitting) sets are disjoint (split). So $\mathcal{F}_{n+1}$ is special, and $\bigcup \mathcal{F}_n = A_1 \cap \cdots \cap A_n = A_n$. □

Definition 7.7 (special functions)**.** Let $0 < \beta < \alpha < \theta < \pi/2$ and $\mathcal{F}$ special. A *pair of special α, β, θ functions* for $\mathcal{F}$ is a pair (ε, δ) of positive functions on $E(\mathcal{F})$ and $\mathcal{F}^2$ such that: (i) for each $\eta > 0$ only finitely many $F \in \mathcal{F}^2$ have $\delta(F) \ge \eta$; (ii) for each $\eta > 0$ only finitely many $x \in E(\mathcal{F})$ have $\varepsilon(x) \ge \eta$; (iii) the sets $s(x, \varepsilon(x), \theta)$, $x \in E(\mathcal{F})$, are pairwise disjoint; (iv) the sets $B(F, \delta(F), \alpha, \beta)$, $F \in \mathcal{F}^2$, are pairwise disjoint; (v) $s(x, \varepsilon(x), \theta) \cap B(F, \delta(F), \alpha, \beta) = \emptyset$ for $x \in E(\mathcal{F})$, $F \in \mathcal{F}^2$.

Lemma 7.8 ([6, (XX)]; [3, Lemma 13])**.** *Every special family has a pair of special α, β, θ functions.*

Proof. Enumerate $\mathcal{F} = \{F_1, F_2, \dots\}$; for $F \in \mathcal{F}$ let $k(F)$ be its least index, and for $e \in E(\mathcal{F})$ let $k(e)$ be the least k such that e is an endpoint of $J(F_k)$. Let $E(k)$ be the set of endpoints of the

$J(F_j)$, $j \le k$, $F_j \neq \emptyset$. Put $c = \cot\theta + \cot\beta$ and define

$$\delta(F) = \min\Big(\Big\{\tfrac{1}{k(F)}\Big\} \cup \Big\{\tfrac{\operatorname{dist}(F_j,F)}{2\cot\beta} : j < k(F),\ F_j \in \mathcal{F}^2,\ F_j \cap F = \emptyset\Big\} \cup \Big\{\tfrac{\operatorname{dist}(e,F)}{c} : e \in E(k(F)-1) \cap J(F)^* \setminus F\Big\}\Big)$$

$$\varepsilon(e) = \min\Big(\Big\{\tfrac{1}{k(e)}\Big\} \cup \Big\{\tfrac{|e-e'|}{2\cot\theta} : e' \in E(k(e)) \setminus \{e\}\Big\} \cup \Big\{\tfrac{\operatorname{dist}(e,F_j)}{c} : j \le k(e),\ F_j \in \mathcal{F}^2,\ e \in J(F_j)^* \setminus F_j\Big\}\Big).$$

These are minima of finite sets of positive numbers. Condition (i): $\delta(F) \ge \eta$ forces $k(F) \le 1/\eta$; (ii): $\varepsilon(e) \ge \eta$ forces $k(e) \le 1/\eta$, and each index contributes at most two endpoints. (iii) If $e \neq e'$, say $k(e') \le k(e)$, then $e' \in E(k(e))$ and (R4) applies. (iv) If $F \neq F'$ in $\mathcal{F}^2$ split, use (R5); if they are disjoint, say $k(F') < k(F)$, use (R6). (v) Let $e \in E(\mathcal{F})$ and $F \in \mathcal{F}^2$. If $e \notin J(F)^*$ use (R8) ($\beta < \alpha < \theta$). Otherwise $e \notin F$: indeed e is an endpoint of some $J(F')$, $e \in F'$, and $F' \neq F$; if $e \in F$ then F, F' would split at e, making e an endpoint of $J(F)$. If $k(F) \le k(e)$ the definition of $\varepsilon(e)$ and (R9) apply; if $k(F) > k(e)$ then $e \in E(k(F)-1)$ and the definition of $\delta(F)$ and (R9) apply. □

Lemma 7.9 ([6, (XXI)]; [3, Lemma 14])**.** *If (ε,δ) is a special pair and $0 < \varepsilon_1 \le \varepsilon$, $0 < \delta_1 \le \delta$, then (ε_1,δ_1) is a special pair.*

Proof. The regions shrink as the parameters ε, δ decrease. □

7.3. **The function g.**

Theorem 7.10 ([6, (A)]; [3, Theorem 4, proof])**.** *Let $A \subseteq X$ be $F_{\sigma\delta}$. There is a continuous $g : \mathbb{H} \to [0,1]$ such that (a) for each $x \in A$, $g(z) \to 0$ as $z \to x$ through $s(x, 1, 3\pi/8)$; and (b) if $x \in \mathbb{R}$ and $g \to 0$ along some arc at x in $\mathbb{H}$, then $x \in A$.*

Proof. Take $\mathcal{F}_n$ as in Lemma 7.6 and $E_n = E(\mathcal{F}_n)$. Fix sequences $\beta_n \uparrow \pi/4$ in $(0,\pi/4)$, $\alpha_n \downarrow \pi/4$ in $(\pi/4,\pi/3)$, $\theta_n \uparrow 3\pi/8$ in $(\pi/3, 3\pi/8)$, all strictly monotone.

Claim. If $x \in E_{n+1} \setminus E_n$, there is a unique $F \in \mathcal{F}_n^2$ with $x \in F \cap J(F)^*$; and no $x \in E_n$ lies in $F \cap J(F)^*$ for any $F \in \mathcal{F}_n^2$. Indeed $x \in K \subseteq F$ for some $K \in \mathcal{F}_{n+1}$, $F \in \mathcal{F}_n$; if x were an endpoint of $J(F)$ it would be in E_n; uniqueness and the last assertion follow from the remark after Definition 7.4. Likewise each $K \in \mathcal{F}_{n+1}^2$ lies in a unique $F \in \mathcal{F}_n$, and $F \in \mathcal{F}_n^2$.

Construction. Let $(\varepsilon(1,\cdot),\delta(1,\cdot))$ be any special $\alpha_1,\beta_1,\theta_1$ pair for $\mathcal{F}_1$. Given $(\varepsilon(n,\cdot),\delta(n,\cdot))$, take any special $\alpha_{n+1},\beta_{n+1},\theta_{n+1}$ pair (ε,δ) for $\mathcal{F}_{n+1}$ and set: for $x \in E_{n+1}\setminus E_n$, with F from the Claim, $\varepsilon(n+1,x) = \min\{\varepsilon(x),\zeta(x)\}$, where $\zeta(x) > 0$ is given by (R10) for $B(F,\delta(n,F),\alpha_n,\beta_n)$ and $\theta_{n+1} > \beta_n$; for $x \in E_{n+1} \cap E_n$, $\varepsilon(n+1,x) = \min\{\varepsilon(x), \tfrac12\varepsilon(n,x)\}$; for $K \in \mathcal{F}_{n+1}^2$, $K \subseteq F$, $\delta(n+1,K) = \min\{\delta(K), \tfrac12\delta(n,F)\}$. By Lemma 7.9 this is a special pair, and by (R3), (R7), (R10):

(i) if $x \in E_{n+1}$, $F \in \mathcal{F}_n^2$, $x \in F \cap J(F)^*$, then $\operatorname{Cl} s(x,\varepsilon(n+1,x),\theta_{n+1}) \cap \mathbb{H} \subseteq B(F,\delta(n,F),\alpha_n,\beta_n)$;
(ii) if $x \in E_{n+1} \cap E_n$, then $\operatorname{Cl} s(x,\varepsilon(n+1,x),\theta_{n+1}) \cap \mathbb{H} \subseteq s(x,\varepsilon(n,x),\theta_n)$;
(iii) if $K \in \mathcal{F}_{n+1}^2$, $F \in \mathcal{F}_n^2$, $K \subseteq F$, then $\operatorname{Cl} B(K,\delta(n+1,K),\alpha_{n+1},\beta_{n+1}) \cap \mathbb{H} \subseteq B(F,\delta(n,F),\alpha_n,\beta_n)$.

The sets U_n. Let U_n be the union of the sets $s(x,\varepsilon(n,x),\theta_n)$, $x \in E_n$, and $B(F,\delta(n,F),\alpha_n,\beta_n)$, $F \in \mathcal{F}_n^2$. These pieces are open and pairwise disjoint, so each component W of U_n lies in one piece, and by (R2)

$$\overline{W} \cap \mathbb{R} \subseteq \bigcup \mathcal{F}_n. \tag{7.1}$$

Each piece lies in $\{y < \varepsilon(n,x)\}$ or $\{y < \delta(n,F)\}$, so by (i), (ii) of Definition 7.7 the pieces form a locally finite family in $\mathbb{H}$, and $\operatorname{Cl} U_n \cap \mathbb{H}$ is the union of the closures of the pieces. By (i)–(iii) and the Claim, $\operatorname{Cl} U_{n+1} \cap \mathbb{H} \subseteq U_n$; in particular $U_{n+1} \subseteq U_n$.

The function. Let $g_n : \mathbb{H} \to [0,1]$ be continuous with $g_n = 1$ on $\mathbb{H} \setminus U_n$ and $g_n = 0$ on $\operatorname{Cl} U_{n+1} \cap \mathbb{H}$ (Urysohn), and $g = \sum_n 2^{-n} g_n$. Then g is continuous with values in $[0,1]$, and

$$g \ge 2^{1-n} \text{ on } \mathbb{H} \setminus U_n, \qquad g \le 2^{-n} \text{ on } U_{n+1}. \tag{7.2}$$

(a) Let $x_0 \in A$ and $n \in \mathbb{N}$. Then $x_0 \in F$ for some $F \in \mathcal{F}_{n+1}$. If $x_0 \in E_{n+1}$, put $\eta = \varepsilon(n+1, x_0)$; as $\theta_{n+1} < 3\pi/8$, $s(x_0, \eta, 3\pi/8) \subseteq s(x_0, \varepsilon(n+1, x_0), \theta_{n+1}) \subseteq U_{n+1}$. Otherwise $x_0 \in F \cap J(F)^*$ with $F \in \mathcal{F}^2_{n+1}$, and (R10) with $\beta_{n+1} < 3\pi/8$ gives $\eta > 0$ with $s(x_0, \eta, 3\pi/8) \subseteq B(F, \delta(n+1, F), \alpha_{n+1}, \beta_{n+1}) \subseteq U_{n+1}$. In both cases $g \le 2^{-n}$ on $s(x_0, 1, 3\pi/8) \cap \{y < \eta\}$.

(b) If $g \to 0$ along an arc γ at x, then for each n a subarc γ' at x has $g < 2^{1-n}$ on $\gamma' \setminus \{x\}$, which therefore lies in U_n, hence in one component W; so $x \in \overline{W} \cap \mathbb{R} \subseteq \bigcup \mathcal{F}_n$ by (7.1). Thus $x \in \bigcap_n \bigcup \mathcal{F}_n = A$. □

Theorem 7.11 ([3, Theorem 4])**.** *For every $F_{\sigma\delta}$ set $A \subseteq \mathbb{R}$ there is a bounded continuous $f : \mathbb{H} \to \mathbb{C}$ with $\mathrm{CC}(f) = A$.*

Proof. Let $A' = \Psi_{\mathbb{H}}^{-1}(A) \subseteq X$, an $F_{\sigma\delta}$ set, g as in Theorem 7.10 for A', and $f_0 = g \sin(1/y) + ig$ on Q°. If $x \in A'$, $f_0 \to 0$ along a short vertical segment at x (which lies in $Q^\circ \cap s(x, 1, 3\pi/8)$ apart from x), since $|f_0| \le 2g$. If $f_0 \to c + di$ along an arc γ at x, then $g \to d$; if $d \neq 0$, then since $y \to 0$ continuously along γ, every tail of γ contains points with $\sin(1/y) = 1$ and with $\sin(1/y) = -1$, so $\operatorname{Re} f_0$ has no limit. Hence $d = 0$ and $x \in A'$ by Theorem 7.10(b). So $\mathrm{CC}(f_0) = A'$, and $f = f_0 \circ \Psi_{\mathbb{H}}^{-1}$ has $\mathrm{CC}(f) = A$ by Lemma 4.18. □

7.4. The function h.

Lemma 7.12 ([6, (B)(I)]; [3, Lemma 17])**.** *(a) If $f : (-1, 1) \to \mathbb{R}$ is strictly increasing, there is a continuous weakly increasing $f^* : \mathbb{R} \to [-1, 1]$ with $f^*(f(x)) = x$ for all x. (b) If $f : \mathbb{R} \to \mathbb{R}$ is strictly increasing and $f(\mathbb{R})$ is unbounded above and below, there is a continuous weakly increasing $f^* : \mathbb{R} \to \mathbb{R}$ with $f^* \circ f = \mathrm{id}$.*

Proof. (a) $f^*(t) = \sup(\{-1\} \cup \{x : f(x) \le t\})$. It is weakly increasing, and $\{x : f(x) \le f(x_0)\} = (-1, x_0]$ gives $f^*(f(x_0)) = x_0$. Its range contains $(-1, 1)$, so it is dense in $[-1, 1]$, and a monotone map onto a dense subset of an interval is continuous. (b) The same with $f^*(t) = \sup\{x : f(x) \le t\}$, whose range is $\mathbb{R}$. □

Lemma 7.13 ([6, (B)(II)]; [3, Lemma 16])**.** *Let M be a metric space and $u : M \to \mathbb{R}$. Suppose that whenever $p_n \to p$ in M and $u(p_n) \to y$ in $[-\infty, +\infty]$, we have $y \in \mathbb{R}$ and $y = u(p)$. Then u is continuous.*

Proof. If $u(p_n) \not\to u(p)$, a subsequence converges in $[-\infty, \infty]$ to some $y \neq u(p)$. □

Theorem 7.14 ([6, (B)]; [3, Theorem 6])**.** *Let $E \subseteq X$ and let $\varphi : E \to \mathbb{R}^m$ be of honorary Baire class ≤ 2. There is a continuous $h : \mathbb{H} \to \mathbb{R}^m$ such that for each $p \in E$ there is a segment λ_p at p, contained in $s(p, 1, 3\pi/8) \cup \{p\}$, along which $h \to \varphi(p)$. If φ is bounded, h can be taken bounded.*

Proof. Let $\varphi = \psi$ on $E \setminus N_0$ with N_0 countable and $\psi \in \mathcal{B}_1(E, \mathbb{R}^m)$. Let $N \supseteq N_0$ be a countably infinite subset of X, enumerated injectively as $s_1, s_2, \ldots$, and let $w(s_n) = (\sqrt{2} - 1)2^{-n-1}$, so $\sum_s w(s) < \sqrt{2} - 1 = \cot(3\pi/8)$. Define on X

$$f(x) = x + \sum_{s \in N,\ s < x} w(s),$$

which is strictly increasing with $|f(x) - x| < \sqrt{2} - 1$. Let f^* be as in Lemma 7.12(a).

The function u. For $\langle x, y\rangle$ with $0 < y < 1$ the function $v \mapsto v - f^*\big((x - (1-y)v)/y\big)$ is continuous and strictly increasing, and lies in $[v - 1, v + 1]$; so it has a unique zero $u(x, y) \in [-1, 1]$. If $\langle x_n, y_n\rangle \to \langle x, y\rangle$ and $u(x_n, y_n) \to u_0$, then $u_0 \in [-1, 1]$ and continuity of f^* gives $u_0 = f^*((x - (1-y)u_0)/y)$, so $u_0 = u(x, y)$; by Lemma 7.13, u is continuous.

The function h_0. By Theorem 3.14 there are continuous $g_n : \mathbb{R} \to \mathbb{R}^m$ with $g_n \to \psi$ on E. For $n \geq 2$ and $1/(n+1) \leq y \leq 1/n$ put

$$h_0(x,y) = \big(yn(n+1) - n\big)g_n(u(x,y)) + \big((n+1) - yn(n+1)\big)g_{n+1}(u(x,y)),$$

and $h_0(x,y) = h_0(x, \frac{1}{2})$ for $y \geq \frac{1}{2}$. The two formulas agree at $y = 1/n$, so h_0 is continuous on $\mathbb{H}$; the coefficients lie in $[0,1]$ and add up to 1.

The correction. Let $l_n = f(s_n) = \sup_{x \leq s_n} f(x)$, $r_n = \inf_{x > s_n} f(x) = l_n + w(s_n)$, $z_n = \varphi(s_n) - \psi(s_n)$ if $s_n \in E$ and $z_n = 0$ otherwise, $t_n(x,y) = (x - (1-y)s_n)/y$, and

$$\Lambda_n(x,y) = \big[(1 - ny) \vee 0\big]\Big[\Big(1 - \frac{|r_n + l_n - 2t_n(x,y)|}{r_n - l_n}\Big) \vee 0\Big] z_n.$$

Λ_n is continuous and vanishes for $y \geq 1/n$, and its second factor vanishes unless $l_n < t_n < r_n$. So $h = h_0 + \sum_n \Lambda_n$ is continuous on $\mathbb{H}$.

Points of E. Let $p \in E$ and let λ be the part with $0 < y \leq \frac{1}{2}$ of the line $x = (f(p) - p)y + p$. On λ, $t_n = f(p) + (p - s_n)(1/y - 1)$, which is $\geq f(p) \geq r_n$ if $p > s_n$, is $\leq f(p) < l_n$ if $p < s_n$, and equals l_n if $p = s_n$; so every Λ_n vanishes on λ. Also $(x - (1-y)p)/y = f(p)$ there, so $u = p$, and h_0 lies on the segment from $g_n(p)$ to $g_{n+1}(p)$ when $1/(n+1) \leq y \leq 1/n$. Hence $h \to \psi(p)$ along λ, which is $\varphi(p)$ unless $p \in N_0$.

Points of N_0. Let $p = s_m \in E$, $\mu_m = (r_m + l_m)/2$, and let λ' be the part with $0 < y \leq \frac{1}{2}$ of the line $x = (\mu_m - s_m)y + s_m$. On λ', $t_n = \mu_m + (s_m - s_n)(1/y - 1)$. If $s_n < s_m$ then $t_n > l_m \geq r_n$; if $s_n > s_m$ then $t_n < r_m \leq l_n$; so $\Lambda_n = 0$ for $n \neq m$, while $\Lambda_m = [(1 - my) \vee 0]z_m \to z_m$. For $a < s_m < b$ we have $f(a) \leq l_m < \mu_m < r_m \leq f(b)$, hence $a \leq f^*(\mu_m) \leq b$; so $f^*(\mu_m) = s_m$ and $u = s_m$ on λ'. Thus $h \to \psi(s_m) + z_m = \varphi(s_m)$ along λ'.

The slopes satisfy $|f(p) - p| < \sqrt{2} - 1$ and $0 \leq \mu_m - s_m < \sqrt{2} - 1$, so $\lambda, \lambda' \subseteq s(p, 1, 3\pi/8)$. If φ is bounded, choose b with $\varphi(E) \subseteq (-b, b)^m$ and clamp each coordinate of h to $[-b, b]$; the limits along λ_p are unchanged. □

7.5. **The function k.**

Theorem 7.15 ([6, (C)])**.** *Let $d : (0,1] \to (0,\infty)$ be weakly increasing. There is a continuous $k : \mathbb{R} \times (0,1] \to \mathbb{C}$ with $|k| \leq \sqrt{2}$ such that for every $a \in (0,1]$ and every connected $\gamma \subseteq [-1,1] \times (0,a]$ containing two points at distance $\geq d(a)$, there are $p, q \in \gamma$ with $|k(p) - k(q)| \geq 2$.*

Proof. Let $p(x) = \frac{1}{2}\int_0^x d(t)\,dt$ for $0 < x \leq 1$: positive, continuous, strictly increasing, with $p(x) \leq \frac{1}{2}d(x)$. For $a \in (0,1]$ the function $1/p$ is uniformly continuous on $[a/4, 1]$, so there is $\varepsilon \in (0, a/4]$ such that $(a/2 \leq x_1 \leq 1,\ x_2 \in (0,1],\ |x_1 - x_2| < \varepsilon)$ implies $|p(x_1)^{-1} - p(x_2)^{-1}| \leq \frac{1}{2}$. Let $e(a)$ be the supremum of such ε in $(0,1]$; the implication holds with $\varepsilon = e(a)$, and e is weakly increasing because an ε that works for a works for every $a' \geq a$ (the condition $a'/2 \leq x_1$ is stronger). Let $q(x) = \int_0^x e(t)\,dt$, so q is positive, continuous, increasing and $q \leq e$; let $m = \min\{p, q\}$. Define

$$k_1(y) = \sin\frac{2\pi}{y\,m(y)}, \qquad k_2(x,y) = \sin\frac{4\pi x}{p(y)}, \qquad k = k_1 + ik_2.$$

Let $z_1, z_2 \in \gamma$ with $|z_1 - z_2| \geq d(a)$, $z_j = \langle x_j, y_j\rangle$, $y_2 \leq y_1 \leq a$, and $a' = \min\{a, 2y_1\}$, so $a'/2 \leq y_1 \leq a' \leq a$. As $m(a') \leq \frac{1}{2}d(a') \leq \frac{1}{2}d(a)$, either

$$(12)\ \ |y_1 - y_2| \geq m(a'), \qquad \text{or} \qquad (13)\ \ |y_1 - y_2| < m(a') \text{ and } |x_1 - x_2| \geq \tfrac{1}{2}d(a').$$

In case (12), using $m(y_2) \leq m(y_1) \leq m(a')$ and $y_1, y_2 \leq 1$,

$$\frac{2\pi}{y_2 m(y_2)} - \frac{2\pi}{y_1 m(y_1)} \geq \frac{2\pi(y_1 - y_2)}{y_1 y_2 m(y_1)} \geq \frac{2\pi m(a')}{y_1 y_2 m(y_1)} \geq 2\pi.$$

The continuous function $2\pi/(y\,m(y))$ on the connected set γ therefore takes all values in an interval of length 2π, so k_1 takes the values 1 and -1 on γ. In case (13), $|y_1-y_2|<m(a')\le e(a')$ gives $|p(y_1)^{-1}-p(y_2)^{-1}|\le\frac12$, and $p(y_1)\le p(a')\le\frac12 d(a')$, so with $|x_2|\le 1$

$$\Big|\frac{4\pi x_1}{p(y_1)}-\frac{4\pi x_2}{p(y_2)}\Big|\ge 4\pi\Big[\frac{|x_1-x_2|}{p(y_1)}-\Big|\frac{1}{p(y_1)}-\frac{1}{p(y_2)}\Big|\Big]\ge 4\pi\Big[1-\frac12\Big]=2\pi,$$

and k_2 takes the values ±1 on γ. In either case some $p,q\in\gamma$ have $|k(p)-k(q)|\ge 2$. □

7.6. **Assembly.**

Theorem 7.16 ([6, (D)])**.** *Let $A\subseteq X$ be $F_{\sigma\delta}$ and $\varphi:A\to\mathbb{C}$ bounded and of honorary Baire class ≤ 2. There is a bounded continuous $f:Q\to\mathbb{C}$ such that A is the set of curvilinear convergence of f (arcs in Q°) and φ is a boundary function for f.*

Proof. Let g be given by Theorem 7.10 and h (bounded, with $\mathbb{C}=\mathbb{R}^2$) by Theorem 7.14 for $E=A$, both restricted to Q. For $t\in(0,1]$ let $d_1(t)$ (resp. $d_2(t)$) be the supremum of the $\delta\in(0,1]$ such that $z_1,z_2\in Q$, $\operatorname{Im} z_j\ge t$, $|z_1-z_2|<\delta$ imply $|h(z_1)-h(z_2)|\le t$ (resp. $|g(z_1)-g(z_2)|\le t$); these are positive by uniform continuity on $[-1,1]\times[t,1]$, and weakly increasing. Let $d(t)=\min\{\frac14 d_1(t/2),\frac14 d_2(t/2),t/4\}$, k as in Theorem 7.15, and $f=h+gk$.

If $x\in A$, $h\to\varphi(x)$ along a segment in $s(x,1,3\pi/8)$, along which $gk\to 0$; a short initial piece of this segment lies in Q° (Lemma 4.5). So φ is a boundary function and $A\subseteq \mathrm{CC}(f)$. Conversely let f have a limit along an arc γ at $x\in X$ in Q°; we show $g\to 0$ along γ, so $x\in A$ by Theorem 7.10(b). If not, there are $\varepsilon\in(0,1]$ and $z_n=\langle x_n,y_n\rangle\in\gamma\setminus\{x\}$, $z_n\to x$, with $|g(z_n)|\ge\varepsilon$. Fix n large: $y_n<3\varepsilon/16$, and the other endpoint of γ is farther than y_n from z_n. Let $a=4y_n/3\le 1$. By Lemma 4.5(b) a connected $\gamma'\subseteq\gamma\cap\overline{S(d(a),z_n)}$ contains z_n and a point at distance $d(a)$ from it. Since $d(a)\le a/4=y_n/3$, $\gamma'\subseteq[-1,1]\times[a/2,a]$, and Theorem 7.15 gives $p,q\in\gamma'$ with $|k(p)-k(q)|\ge 2$. As $|p-q|\le 2d(a)<d_1(a/2)$ and $|p-z_n|,|q-z_n|\le d(a)<d_2(a/2)$, all with imaginary parts $\ge a/2$, we get $|h(p)-h(q)|\le a/2$ and $|g(p)-g(z_n)|,|g(q)-g(z_n)|\le a/2<\varepsilon/8$. Hence

$$|f(p)-f(q)|\ge|g(z_n)||k(p)-k(q)|-|k(p)||g(p)-g(z_n)|-|k(q)||g(q)-g(z_n)|-|h(p)-h(q)|>2\varepsilon-\tfrac{2\sqrt2+1}{8}\varepsilon>\varepsilon.$$

Since $|p-z_n|,|q-z_n|\le y_n/3\to 0$, f has no limit along γ, a contradiction. □

Theorem 7.17 ([6, (E)])**.** *Let $A\subseteq C$ be $F_{\sigma\delta}$ and $\varphi:A\to\mathbb{C}$ bounded and of honorary Baire class ≤ 2. There is a bounded continuous $f:\mathbb{D}\to\mathbb{C}$ with $\mathrm{CC}(f)=A$ such that φ is a boundary function for f.*

Proof. If $A=\emptyset$ take $f(z)=\exp\big(i/(1-|z|)\big)$: along any arc, $1/(1-|z|)$ tends to $+\infty$ continuously, so f has no limit. Otherwise rotate so that $1\in A$. Let $A'=\Psi_{\mathbb{D}}^{-1}(A\setminus\{1\})\subseteq X$, which is $F_{\sigma\delta}$, and apply Theorem 7.16 to A' and $\varphi\circ\Psi_{\mathbb{D}}$ (of honorary Baire class ≤ 2 by Lemma 3.23), obtaining f_1; put $f_0=f_1\circ\Psi_{\mathbb{D}}^{-1}$ on G°. On the set G'' of Lemma 4.13(c), which is closed in $\mathbb{D}$, f_0 is continuous and bounded; set $F=f_0$ on G'' and $F=\varphi(1)$ on the radius $[0,1)$, which is closed in $\mathbb{D}$ and disjoint from G''. By the Tietze extension theorem (Fact 4.7) F extends to a bounded continuous f on $\mathbb{D}$. Along the radius $f\to\varphi(1)$. Each $x\in C\setminus\{1\}$ has a neighbourhood W with $W\cap\mathbb{D}\subseteq G''$, so by Lemmas 4.19 and 4.18 the arc limits of f at x are those of f_1 at $\Psi_{\mathbb{D}}^{-1}(x)$. □

Theorem 7.18 ([6, (F)]; McMillan's problem)**.** *Let $A\subseteq C$ be $F_{\sigma\delta}$ and $\varphi:A\to\mathbb{S}^2$ of honorary Baire class ≤ 2. There is a continuous $f:\mathbb{D}\to\mathbb{S}^2$ with $\mathrm{CC}(f)=A$ such that φ is a boundary function for f.*

Proof. Let $\varphi=\psi$ off a countable N with $\psi\in\mathcal{B}_1(A,\mathbb{S}^2)$, and $A_1=A\setminus N$. By Lemmas 3.7 and 3.19, applied in the metric space A_1, there is $\psi_0:A_1\to Z_U\cup Z_V$ with $\Phi\circ\psi_0=\varphi|_{A_1}$ and

$\sigma\circ\psi_0$ of class F_σ. For $x\in N$ choose $\psi_0(x)\in Z_U\cup Z_V$ with $\Phi(\psi_0(x))=\varphi(x)$ (possible since $\Phi(Z_U\cup Z_V)=\mathbb{S}^2$). Then $\sigma\circ\psi_0: A\to\mathbb{R}^2$ is bounded (Lemma 3.18(v)) and of honorary Baire class ≤ 2 (Theorem 3.22). By Theorem 7.17 there is a bounded continuous $f_1:\mathbb{D}\to\mathbb{C}$ with $\mathrm{CC}(f_1)=A$ and boundary function $\sigma\circ\psi_0$. Put $f=\Phi\circ\sigma^{-1}\circ f_1$. Then $f\to\Phi(\psi_0(x))=\varphi(x)$ along the arcs at $x\in A$. If $x\notin A$ and γ is an arc at x, f_1 has no limit along γ, so its cluster set (with values in a closed disk K containing $f_1(\mathbb{D})$) is a connected set with more than one point (Lemma 4.6), hence infinite; by Lemma 4.6 with $\Theta=\Phi\circ\sigma^{-1}|_K$, $C(f,\gamma)$ is its image under Θ, which is at most three-to-one, so $C(f,\gamma)$ is infinite and f has no limit along γ. □

Theorem 7.19 (half-plane versions; [3, Theorems 4, 6, 7]). *(a) If $A\subseteq\mathbb{R}$ is $F_{\sigma\delta}$ and $\varphi: A\to\mathbb{C}$ is bounded and of honorary Baire class ≤ 2, there is a bounded continuous $f:\mathbb{H}\to\mathbb{C}$ with $\mathrm{CC}(f)=A$ and boundary function φ. The same holds with $\mathbb{S}^2$ in place of $\mathbb{C}$ (and f continuous).*

(b) ([3, Theorem 6]) If $E\subseteq\mathbb{R}$ is arbitrary and $\varphi: E\to\mathbb{R}^m$ is of honorary Baire class ≤ 2, there is a continuous $f:\mathbb{H}\to\mathbb{R}^m$ having φ as a boundary function.

(c) ([3, Theorem 7]) The same with $\mathbb{S}^2$ in place of $\mathbb{R}^m$.

Proof. (a) Transfer Theorem 7.16 by $\Psi_{\mathbb{H}}$ (Lemma 4.18); no exceptional point occurs. For $\mathbb{S}^2$ argue as in Theorem 7.18. (b) Apply Theorem 7.14 to $\Psi_{\mathbb{H}}^{-1}(E)$ and $\varphi\circ\Psi_{\mathbb{H}}$; the segments have tails in Q°; transfer. (c) Write $\varphi=\Phi\circ\sigma^{-1}\circ\chi$ with $\chi: E\to\mathbb{R}^2$ bounded and of honorary Baire class ≤ 2, exactly as in Theorem 7.18, apply (b) to χ, and compose with $\Phi\circ\sigma^{-1}$. □

8. Boundary functions of discontinuous functions

Sources: [3, Chapter II]; [1, Theorems 4–6]. We work in $\mathbb{H}$ with boundary $\mathbb{R}$.

8.1. Baire functions.

Definition 8.1. For arcs γ_x,γ_y at $x,y\in\mathbb{R}$ and $n\in\mathbb{N}$, γ_x *meets* γ_y *in* $\overline{\mathbb{H}}_n$ if there are subarcs $\gamma_x'\ni x$ and $\gamma_y'\ni y$ of them, contained in $\overline{\mathbb{H}}_n$, with $\gamma_x'\cap\gamma_y'\neq\emptyset$. If γ_x meets γ_y in $\overline{\mathbb{H}}_n$ and $n'\le n$, it meets γ_y in $\overline{\mathbb{H}}_{n'}$.

Lemma 8.2 (the "Figure 5" argument of [3, proof of Theorem 8]). *Let $x<t<y$ be real, let α,β be arcs at x,y contained in $\overline{\mathbb{H}}_n$ with $\alpha\cap\beta\neq\emptyset$, and let τ be an arc at t, contained in $\overline{\mathbb{H}}_n$, whose other endpoint lies on X_n. Then τ meets $\alpha\cup\beta$.*

Proof. By Fact 4.3, $\alpha\cup\beta$ contains a simple arc σ from x to y, a crosscut of $\mathbb{H}$ contained in $\overline{\mathbb{H}}_n$. Suppose $\tau\cap\sigma=\emptyset$. Let V,U be the bounded and unbounded components of $\mathbb{H}\setminus\sigma$ (Lemma 4.14). A small half-disk $S(r,t)\cap\mathbb{H}$ misses σ, is connected, and meets V because $t\in(x,y)\subseteq\overline{V}$; so it lies in V, and therefore so does the connected set $\tau\setminus\{t\}$. Its endpoint q on X_n is then in the open set V, so $q+i\varepsilon\in V$ for small $\varepsilon>0$; but $\{\mathrm{Im}\,z>1/n\}\subseteq U$ since $\max_\sigma\mathrm{Im}\le 1/n$. Contradiction. □

Theorem 8.3 ([3, Theorem 8]; [1, Theorem 4]). *Let Y be a separable arcwise connected metric space, $\xi\ge 1$, $f:\mathbb{H}\to Y$ of Baire class ξ, $E\subseteq\mathbb{R}$, and $\varphi: E\to Y$ a boundary function for f. Then φ is of Baire class $\xi+1$.*

Proof. Let $U\subseteq Y$ be open, $V=Y\setminus\overline{U}$, $A=\varphi^{-1}(U)$, $B=\varphi^{-1}(V)$, $P=A\cup B$. For $x\in P$ choose (Lemma 4.5) an arc γ_x at x with $f\to\varphi(x)$ along it, $\gamma_x\subseteq\{|z-x|\le 1\}$, and $\gamma_x\setminus\{x\}\subseteq f^{-1}(U)$ if $x\in A$, $\subseteq f^{-1}(V)$ if $x\in B$. If $x\in A$ and $y\in B$ then $\gamma_x\cap\gamma_y=\emptyset$. Let

$$L_a=\{x\in A:\forall n\ \exists y\in P\setminus\{x\},\ \gamma_y\text{ meets }\gamma_x\text{ in }\overline{\mathbb{H}}_n\},\qquad M_a=A\setminus L_a,$$
$$L_b=\{x\in B:\forall n\ \exists y\in P\setminus\{x\},\ \gamma_y\text{ meets }\gamma_x\text{ in }\overline{\mathbb{H}}_n\},\qquad M_b=B\setminus L_b,$$

$L = L_a \cup L_b$, $M = M_a \cup M_b$. For $x \in M$ let $n(x)$ be such that γ_x meets no γ_y ($y \neq x$) in $\overline{\mathbb{H}}_{n(x)}$; the same then holds for all $n \geq n(x)$. Let K_n be the set of $x \in P$ such that γ_x meets X_n and, if $x \in M$, $n \geq n(x)$. Then $K_n \subseteq K_{n+1}$ and $P = \bigcup_n K_n$.

Step 1: $L_a \approx G_a \cap P$ for a G_δ set G_a, and $L_b \approx G_b \cap P$ for a G_δ set G_b. Let $x \in L_a$ and $n \in \mathbb{N}$; choose $y \in P \setminus \{x\}$ with γ_y meeting γ_x in $\overline{\mathbb{H}}_n$, and let I_x^n be the closed interval with endpoints x, y. We claim $I_x^n \subseteq L_a \cup (\mathbb{R} \setminus K_n)$. Let $t \in I_x^n \cap K_n$. If $t \neq x, y$, let τ be the subarc of γ_t from t to its first point on X_n; by Lemma 8.2, γ_t meets γ_x or γ_y in $\overline{\mathbb{H}}_n$. The same holds trivially for $t = y$. If $t \in M$ this contradicts $n \geq n(t)$; so $t \notin M$. Since γ_x meets γ_y and $x \in A$, $y \notin B$, so $y \in A$; likewise $t \in A$. Hence $t \in A \setminus M = L_a$. Put $W_n = \bigcup_{x \in L_a} I_x^n$. Then $L_a \subseteq W_n \cap P \subseteq (L_a \cup (\mathbb{R} \setminus K_n)) \cap P$, so $L_a = \bigcap_n (W_n \cap P)$. By Lemma 3.34 each W_n is $\approx$ an open set O_n; let $G_a = \bigcap_n O_n$. By Lemma 3.33, $L_a \approx G_a \cap P$. The same argument applies to L_b.

Step 2: $M_a \approx F \cap P$ with $F \in P^{\xi+2}(\mathbb{R})$. For $x \in M \cap K_n$ let $p_n(x)$ be the first point of X_n reached along γ_x from x, and $h_n^0(x) = \operatorname{Re} p_n(x)$. If $x < x'$ in $M \cap K_n$, the subarcs from x to $p_n(x)$ and from x' to $p_n(x')$ are disjoint (as $n \geq n(x)$), so $h_n^0(x) < h_n^0(x')$ by Lemma 4.15. Also $|x - h_n^0(x)| \leq 1$. By Lemma 3.36 extend h_n^0 to a weakly increasing $h_n : \mathbb{R} \to \mathbb{R}$ (take $h_n = \mathrm{id}$ if $M \cap K_n = \emptyset$), and let $g_n(x) = f(h_n(x) + i/n)$. By Lemmas 3.29 and 3.37 there is a countable N such that each $g_n|_{M \setminus N}$ is of Baire class ξ. For $x \in M$ and n large, $x \in M \cap K_n$ and $g_n(x) = f(p_n(x))$ with $p_n(x) \in \gamma_x$, $\operatorname{Im} p_n(x) = 1/n$; so $p_n(x) \to x$ along γ_x and $g_n(x) \to \varphi(x)$. Hence $\varphi|_{M \setminus N}$ is of Baire class $\xi + 1$, hence of class $P^{\xi+2}$ (Lemma 3.30), so $A \cap (M \setminus N) = D \cap (M \setminus N)$ for some $D \in P^{\xi+2}(\mathbb{R})$ (Lemma 3.28(f)). Then

$$M_a = A \cap M \approx D \cap M = D \cap (P \setminus L) \approx D \cap \big(\mathbb{R} \setminus (G_a \cup G_b)\big) \cap P =: F \cap P,$$

and $F \in P^{\xi+2}(\mathbb{R})$ because $\mathbb{R} \setminus (G_a \cup G_b)$ is F_σ, i.e. in $P^2 \subseteq P^{\xi+2}$.

Step 3. $A = L_a \cup M_a \approx (G_a \cup F) \cap P$, and $G_a \in Q^2 \subseteq P^{\xi+2}$ since $\xi \geq 1$. Correcting by countable sets (countable sets are in P^2 and co-countable sets in Q^2), $A = T \cap P$ for some $T \in P^{\xi+2}(\mathbb{R})$. Then $T \cap B = A \cap B = \emptyset$, so $\varphi^{-1}(U) = A \subseteq T \cap E \subseteq E \setminus B = \varphi^{-1}(\overline{U})$, and $T \cap E \in P^{\xi+2}(E)$. By Lemma 3.38 (with $\xi + 1 \geq 2$), φ is of Baire class $\xi + 1$. □

Corollary 8.4 ([3, Theorem 8, Corollary]; [1, Theorems 4, 5])**.** *Let Y be a separable, arcwise connected metric space, $f : \mathbb{H} \to Y$ Borel measurable, and $\varphi : E \to Y$ a boundary function for f. Then φ is Borel measurable. In particular ([1, Theorems 4, 5]), if f is a real function of Baire class $\xi \geq 1$ in the upper half-plane, every finite-valued boundary function of f is of Baire class $\xi + 1$, and if f is Borel measurable so is every finite-valued boundary function.*

Proof. By Fact 3.32, f is of some Baire class $\xi < \omega_1$ (if $\xi = 0$, also of class 1); by Theorem 8.3, φ is of Baire class $\xi + 1$, hence Borel measurable (Lemma 3.30). $\mathbb{R}$ is separable and arcwise connected. □

8.2. **Lebesgue measure.** $X_0 = \mathbb{R} \times \{0\}$, $X_1 = \mathbb{R} \times \{1\}$; m is planar Lebesgue measure, m^1 linear measure on a line, and m_e, m_e^1 the outer measures. A *segment* below means a closed line segment with one endpoint $\langle a_0(s), 0\rangle \in X_0$ and the other $\langle a_1(s), 1\rangle \in X_1$; write $\ell_s(y) = (1 - y)a_0(s) + y a_1(s)$.

Definition 8.5 (trapezoids)**.** For $a_0 \leq b_0$, $a_1 \leq b_1$ in $[-\infty, +\infty]$ with $a_i < +\infty$ and $b_i > -\infty$ (and $0 \cdot (\pm\infty) = 0$), let $T(a_0, b_0; a_1, b_1) = \{\langle x, y\rangle : 0 \leq y \leq 1,\ x \in \mathbb{R},\ (1 - y)a_0 + y a_1 \leq x \leq (1 - y)b_0 + y b_1\}$, a *closed trapezoid*; $\emptyset$ is also one. S is a *trapezoid* if $T^\circ \subseteq S \subseteq T$ for a closed trapezoid T, where T° is the interior of T *relative to the strip* $\{0 \leq y \leq 1\}$. For disjoint segments s, s' with $a_i(s) \leq a_i(s')$, $T(s, s') = T(s', s) = T(a_0(s), a_0(s'); a_1(s), a_1(s'))$; and $T(s, s) = s$.

A trapezoid S is Lebesgue measurable and $m(S) = \frac{1}{2}\big[(b_0 - a_0) + (b_1 - a_1)\big] = \frac{1}{2}\big[m^1(S \cap X_0) + m^1(S \cap X_1)\big]$: integrate the lengths $(1 - y)(b_0 - a_0) + y(b_1 - a_1)$ of the horizontal sections, and

note that $S \cap X_i$ lies between (a_i, b_i) and $[a_i, b_i]$. (The restrictions $a_i < +\infty$, $b_i > -\infty$ exclude the degenerate sides $a_i = b_i = \pm\infty$, for which the formula fails.) Two disjoint segments $s \neq s'$ satisfy either $\ell_s < \ell_{s'}$ on $[0,1]$ or $\ell_s > \ell_{s'}$ on $[0,1]$, since $\ell_s - \ell_{s'}$ is affine and does not vanish.

Lemma 8.6 ([3, Lemma 23]). *Let s_1, s_2, s_3, s_4 be segments, any two equal or disjoint. If $T(s_1, s_2) \cap T(s_3, s_4) \neq \emptyset$ then $T(s_1, s_3) \subseteq T(s_1, s_2) \cup T(s_3, s_4)$.*

Proof. The horizontal section of $T(s_i, s_j)$ at height y is the interval between $\ell_{s_i}(y)$ and $\ell_{s_j}(y)$. The order of the four numbers $\ell_{s_i}(y)$ (with ties exactly between equal segments) does not depend on y. Whether two such intervals meet depends only on this order; so if the two sections meet at one height they meet at every height. The union of two meeting intervals is an interval, and it contains $\ell_{s_1}(y)$ and $\ell_{s_3}(y)$. □

Lemma 8.7 ([3, Lemma 24]). *If Σ is a set of pairwise disjoint segments, $\bigcup_{s,s' \in \Sigma} T(s, s')$ is a trapezoid.*

Proof. Let $A_i = \inf_s a_i(s)$, $B_i = \sup_s a_i(s)$ and $T = T(A_0, B_0; A_1, B_1)$. The union lies in T. The segments are totally ordered and both a_0 and a_1 increase along the order; so a sequence s_k descending in this order with $a_0(s_k) \to A_0$ also has $a_1(s_k) \to A_1$ (given s with $a_1(s)$ close to A_1, eventually $s_k \preceq s$), whence $\inf_s \ell_s(y) = (1-y)A_0 + yA_1$, and similarly for the supremum. Hence every $\langle x, y\rangle \in T^\circ$ satisfies $\ell_s(y) < x < \ell_{s'}(y)$ for some $s, s' \in \Sigma$, so it lies in $T(s, s')$. □

Theorem 8.8 ([3, Theorem 9]). *Let Σ be a set of pairwise disjoint segments and $S = \bigcup \Sigma$. Then $m_e(S) = \frac{1}{2}\big[m_e^1(S \cap X_0) + m_e^1(S \cap X_1)\big]$.*

Proof. We may assume $\Sigma \neq \emptyset$. Let $\varepsilon > 0$ and $E_i = S \cap X_i$. Choose open $U \supseteq S$ with $m(U) \leq m_e(S) + \varepsilon$ and relatively open $G_i \supseteq E_i$ in X_i with $m^1(G_i) \leq m_e^1(E_i) + \varepsilon$ (when these are finite). Let $V = \{\langle (1-y)u_0 + yu_1, y\rangle : u_0 \in G_0, u_1 \in G_1, y \in \mathbb{R}\}$. V is open (if $\langle x, y\rangle = \langle (1-y)u_0 + yu_1, y\rangle$, then nearby points are $\langle (1-y')(u_0+t) + y'(u_1+t), y'\rangle$ with small t), $S \subseteq V$ and $V \cap X_i = G_i$. Put $W = U \cap V$.

For $s, s' \in \Sigma$ write $s \equiv s'$ if $T(s, s') \subseteq W$. By Lemma 8.6 (with $T(s_1, s_2) \cap T(s_2, s_3) \supseteq s_2$) this is an equivalence relation; let Γ be the set of classes. Γ *is countable:* for $s \in \Sigma$, s is compact and in W, so there is $\delta_s > 0$ such that $N_s = \{\langle x, y\rangle : 0 \leq y \leq 1,\ |x - \ell_s(y)| < \delta_s\} \subseteq W$. Let $J_i(s) = (a_i(s) - \delta_s, a_i(s) + \delta_s)$. If $J_0(s) \cap J_0(s') \neq \emptyset$ and $J_1(s) \cap J_1(s') \neq \emptyset$, then $|\ell_s(y) - \ell_{s'}(y)| < \delta_s + \delta_{s'}$ for all y, so $T(s, s') \subseteq N_s \cup N_{s'} \subseteq W$ and $s \equiv s'$. Choosing $s(C) \in C$ for $C \in \Gamma$, the open rectangles $J_0(s(C)) \times J_1(s(C))$ are pairwise disjoint, so Γ is countable.

For $C \in \Gamma$ let $T(C) = \bigcup_{s,s' \in C} T(s, s')$, a trapezoid by Lemma 8.7, with $C \subseteq T(C) \subseteq W$. If $C_1 \neq C_2$ then $T(C_1) \cap T(C_2) = \emptyset$: otherwise some $T(s_1, s_1') \cap T(s_2, s_2') \neq \emptyset$ and Lemma 8.6 gives $T(s_1, s_2) \subseteq W$, i.e. $s_1 \equiv s_2$. Let $K_i(C) = T(C) \cap X_i$; these are intervals, pairwise disjoint for fixed i, and $E_i \subseteq \bigcup_C K_i(C) \subseteq W \cap X_i \subseteq G_i$. Put

$$a = m\Big(\bigcup_C T(C)\Big) = \sum_C m(T(C)) = \tfrac{1}{2}\Big[m^1\Big(\bigcup_C K_0(C)\Big) + m^1\Big(\bigcup_C K_1(C)\Big)\Big].$$

Since $S \subseteq \bigcup_C T(C) \subseteq W \subseteq U$, $m_e(S) \leq a \leq m_e(S) + \varepsilon$; and $\frac{1}{2}[m_e^1(E_0) + m_e^1(E_1)] \leq a \leq \frac{1}{2}[m^1(G_0) + m^1(G_1)] \leq \frac{1}{2}[m_e^1(E_0) + m_e^1(E_1)] + \varepsilon$. Let $\varepsilon \to 0$ (the infinite cases are covered by the same inequalities). □

Proposition 8.9 ([3, example after Theorem 9]). *Disjointness cannot be omitted in Theorem 8.8: there are sets $M_0 \subseteq X_0$, $M_1 \subseteq X_1$ of measure zero such that every point of $\{0 < y < 1\}$ lies on a segment from M_0 to M_1.*

Proof. Let M_i be dense G_δ subsets of measure zero, e.g. $\bigcap_n \bigcup_k (q_k - 2^{-n-k}, q_k + 2^{-n-k})$ for an enumeration q_k of $\mathbb{Q}$. Given $\langle x_0, y_0\rangle$ with $0 < y_0 < 1$, the maps $F_0(\theta) = x_0 - y_0 \cot\theta$ and

$F_1(\theta) = x_0 + (1 - y_0)\cot\theta$ are homeomorphisms of $(0,\pi)$ onto $\mathbb{R}$, so $F_0^{-1}(M_0) \cap F_1^{-1}(M_1)$ is a nonempty residual set by Baire's theorem; for θ in it, the line through $\langle x_0, y_0\rangle$ with inclination θ meets X_0 in M_0 and X_1 in M_1. □

Lemma 8.10 ([3, Lemma 25]). *For every $\varepsilon > 0$ there is a strictly increasing $h : \mathbb{R} \to \mathbb{R}$ with $m^1(h(\mathbb{R})) = 0$ and $|x - h(x)| \le \varepsilon$ for all x.*

Proof. For $t \in [0,1)$ with binary digits $t = \sum_i a_i 2^{-i}$ (not ending in 1's) put $c(t) = \sum_i 2a_i 3^{-i}$. Then c maps $[0,1)$ into the Cantor set, which is null, and is strictly increasing: if $t < t'$ first differ at digit k, then $c(t) < \sum_{i<k} 2a_i 3^{-i} + 3^{-k} < c(t')$, the first inequality being strict because the digits of t do not end in 1's. Put $h(x) = n\varepsilon + \varepsilon c(x/\varepsilon - n)$ for $x \in [n\varepsilon, (n+1)\varepsilon)$, $n \in \mathbb{Z}$. □

Theorem 8.11 ([3, Theorem 10]). *There is a family $(\gamma_x)_{x\in\mathbb{R}}$ of arcs, γ_x an arc at x, pairwise disjoint, whose union has planar measure zero.*

Proof. Choose by Lemma 8.10 strictly increasing h_n with $h_n(\mathbb{R})$ null and $|x - h_n(x)| \le 1/n$. For $x \in \mathbb{R}$ let $X_x : [0,1] \to \mathbb{R}$ be $X_x(0) = x$ and, on $[1/(n+1), 1/n]$, the affine function with $X_x(1/(n+1)) = h_{n+1}(x)$ and $X_x(1/n) = h_n(x)$. It is continuous (at 0 because $|X_x(y) - x| \le 1/n$ there), so $\gamma_x = \{\langle X_x(y), y\rangle : 0 \le y \le 1\}$ is an arc at x. If $x < x'$ then $X_x < X_{x'}$ everywhere, so the arcs are disjoint. Let S_n be the union over x of the segments $\{\langle X_x(y), y\rangle : 1/(n+1) \le y \le 1/n\}$. After the affine change of variable $y \mapsto n(n+1)(y - 1/(n+1))$, which multiplies areas by $n(n+1)$, Theorem 8.8 applies; the end sets lie in $h_{n+1}(\mathbb{R})$ and $h_n(\mathbb{R})$, so $m_e(S_n) = 0$. Hence $\bigcup_x \gamma_x \subseteq X_0 \cup \bigcup_n S_n$ is null. □

Corollary 8.12 ([3, Theorem 10, Corollary 1]; [1, Theorem 6]). *Let Y be a metric space with a point 0 and $\varphi : \mathbb{R} \to Y$ arbitrary. There is $f : \mathbb{H} \to Y$ with $f = 0$ outside a set of measure zero such that φ is a boundary function for f.*

Proof. Put $f = \varphi(x)$ on $\gamma_x \setminus \{x\}$ and $f = 0$ elsewhere; f is constant along each γ_x. □

Fact 8.13 (non-measurable sets). *There is a subset of $\mathbb{R}$ that is not Lebesgue measurable.*

Corollary 8.14 ([3, Theorem 10, Corollary 2]). *There is a Lebesgue-measurable $f : \mathbb{H} \to \mathbb{R}$ with a boundary function on $\mathbb{R}$ that is not Lebesgue measurable.*

Proof. Apply Corollary 8.12 to the indicator function of a non-measurable set; $f = 0$ almost everywhere. □

9. Bounded harmonic functions

Source: [5]. A function u defined in $\mathbb{D}$ has the *asymptotic value* a at $\zeta \in C$ if $u \to a$ along some arc at ζ, and the *general limit* a at ζ if $u(z) \to a$ as $z \to \zeta$, $z \in \mathbb{D}$. $P_r(\theta) = (1-r^2)/(1-2r\cos\theta + r^2)$ is the Poisson kernel, $\Delta(S,T) = \inf\{|s - t + 2m\pi| : s \in S, t \in T, m \in \mathbb{Z}\}$ for $S, T \subseteq \mathbb{R}$, and $\operatorname{expi} S = \{e^{it} : t \in S\}$.

Fact 9.1 (Poisson integrals [14, pp. 32–38]). *(i) If φ is a bounded measurable real function on C, its Poisson integral $P[\varphi](re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \varphi(e^{it}) P_r(\theta - t)\,dt$ is a bounded harmonic function in $\mathbb{D}$ with $|P[\varphi]| \le \sup|\varphi|$. (ii) If $\varphi \in L^1(C)$ is defined on a set S of full measure, $\zeta_0 \in S$, and $\varphi|_S$ is continuous at ζ_0, then $P[\varphi]$ has general limit $\varphi(\zeta_0)$ at ζ_0. (iii) If u is harmonic in $\mathbb{D}$ and $\{\int_{-\pi}^{\pi} |u(re^{i\theta})|^p d\theta : 0 \le r < 1\}$ is bounded for some $p > 1$, then the radial limit of u exists at almost every point of C, the function ψ so defined almost everywhere is in $L^1(C)$, and $u = P[\psi]$.*

Fact 9.2 (Jordan curve theorem and accessibility). *If $\Gamma \subseteq \mathbb{C}$ is a Jordan curve, $\mathbb{C} \setminus \Gamma$ has exactly two components, a bounded one B and an unbounded one, each with frontier Γ, and every point $p \in \Gamma$ is accessible from B: there is a simple arc γ with endpoint p and $\gamma \setminus \{p\} \subseteq B$.*

Lemma 9.3 ([5, Lemma])**.** *Let (I_n) be pairwise disjoint nondegenerate closed arcs of C and (Ω_n) Jordan domains in $\mathbb{D}$ with $\overline{\Omega_n}\cap C = I_n$, $\overline{\Omega_n}\cap\overline{\Omega_m}=\emptyset$ for $n\neq m$, and $\operatorname{diam}\Omega_n\to 0$. Then every point of $C\setminus\bigcup_n I_n^*$ is accessible by a simple arc in $\mathbb{D}\setminus\bigcup_n\overline{\Omega_n}$.*

Proof. $\partial\Omega_n$ is a Jordan curve containing I_n; let $\Gamma_n=\partial\Omega_n\setminus I_n^*$, an arc with the endpoints of I_n, and $\phi_n : I_n\to\Gamma_n$ a homeomorphism fixing the endpoints. Let $\Gamma=(C\setminus\bigcup_n I_n^*)\cup\bigcup_n\Gamma_n$ and define $\phi : C\to\Gamma$ by $\phi=\phi_n$ on I_n and $\phi=\mathrm{id}$ elsewhere; let ϕ_p use only $\phi_1,\dots,\phi_p$. Each ϕ_p is continuous, and $|\phi_p-\phi|\le\sup_{n>p}\operatorname{diam}\Omega_n\to 0$, so ϕ is continuous; it is injective and C is compact, so Γ is a Jordan curve. Each Ω_n misses Γ; a small disk about an interior point of I_n misses Γ and meets both Ω_n and $\mathbb{C}\setminus\overline{\mathbb{D}}$, which lies in the unbounded component; so Ω_n lies in the unbounded component. The points of $C\setminus\Gamma=\bigcup I_n^*$ also lie in the unbounded component (they are limits of points outside $\overline{\mathbb{D}}$ near which Γ is absent), and $\overline{\Omega_n}\subseteq\Omega_n\cup\Gamma_n\cup I_n$. Hence the bounded component B (Fact 9.2) lies in $\mathbb{D}\setminus\bigcup_n\overline{\Omega_n}$, and each point of $C\setminus\bigcup I_n^*\subseteq\Gamma$ is accessible from B. □

Lemma 9.4 (general limits along an arc)**.** *Let $u:\mathbb{D}\to\mathbb{R}$ have general limit c at every point of an open arc J of C. Then u has the asymptotic value c at each endpoint of J.*

Proof. Let $J=\operatorname{expi}(t_0,t_1)$ and $\tau_k=t_0+(t_1-t_0)2^{-k-1}$. By compactness of $\operatorname{expi}[\tau_k,\tau_0]$ there is $\rho_k\in(0,1/k)$ with $|u(re^{it})-c|<1/k$ for $t\in[\tau_k,\tau_0]$ and $1-\rho_k<r<1$; we may take ρ_k decreasing. Let $r(t)$ be continuous on $(t_0,\tau_0]$, affine on each $[\tau_{k+1},\tau_k]$, with $r(\tau_k)=1-\rho_{k+1}/2$. Then $\{r(t)e^{it}\}\cup\{e^{it_0}\}$ is an arc at e^{it_0} along which $u\to c$. □

Lemma 9.5 (the intervals)**.** *Let (p_n) be a countable dense subset of $(-\pi,\pi)$. There are natural numbers $n(1)<n(2)<\cdots$ and reals ε_k such that, with $\alpha(k)=p_{n(k)}-\pi\varepsilon_k^2/2$ and $\beta(k)=p_{n(k)}+\pi\varepsilon_k^2/2$: I. $0<\varepsilon_k\le 2^{-k-1}$; II. $-\pi<\alpha(k)<\beta(k)<\pi$; III. the intervals $[\alpha(k),\beta(k)]$ are pairwise disjoint; IV. $\varepsilon_k\le\big(\Delta([\alpha(h),\beta(h)],[\alpha(k),\beta(k)])/4\big)^4$ for $h<k$; V. $p_h\in\bigcup_{i\le k}[\alpha(i),\beta(i)]$ for $h\le n(k)$.*

Proof. Let $n(1)=1$ and choose ε_1 satisfying I, II. Given the first q terms, let $n(q+1)$ be the least n with $p_n\notin\bigcup_{i\le q}[\alpha(i),\beta(i)]$ (it exists because this finite union of closed intervals of total length $<2\pi$ cannot contain a dense set; and $n(q+1)>n(q)$ by V), and choose $\varepsilon_{q+1}>0$ with I, II and $\frac{\pi}{2}\varepsilon_{q+1}^2+4\varepsilon_{q+1}^{1/4}\le\frac12\min_{k\le q}\Delta(p_{n(q+1)},[\alpha(k),\beta(k)])$. Then $\Delta([\alpha(q+1),\beta(q+1)],[\alpha(k),\beta(k)])\ge 4\varepsilon_{q+1}^{1/4}>0$, which gives III and IV, and V holds by the minimality of $n(q+1)$. □

Lemma 9.6 (estimates)**.** *With the data of Lemma 9.5, let $I_n=\operatorname{expi}[\alpha(n),\beta(n)]$, χ_n its indicator, $f_n=P[\chi_n]$, and $\Omega_n=\{re^{i\theta}:1-\varepsilon_n<r<1,\ \Delta(\theta,[\alpha(n),\beta(n)])<(1-r)^{1/4}\}$. Then Ω_n is a Jordan domain with $\overline{\Omega_n}\cap C=I_n$; $\overline{\Omega_n}\cap\overline{\Omega_m}=\emptyset$ for $n\neq m$; $\operatorname{diam}\Omega_n\to 0$; and*

(1) $0\le f_n\le\varepsilon_n$ on $\mathbb{D}\setminus\Omega_n$;
(2) $f_n(re^{i\theta})\le\varepsilon_n\sqrt{1-r}$ when $r>1-\varepsilon_n$ and $\Delta(\theta,[\alpha(n),\beta(n)])\ge(1-r)^{1/4}$;
(3) $f_n\to 0$ along every arc at a point of C that does not meet Ω_n.

Proof. Ω_n is the image of $(0,1)\times(1-\varepsilon_n,1)$ under $(s,r)\mapsto r\exp\big(i[\alpha(n)-(1-r)^{1/4}+s(\beta(n)-\alpha(n)+2(1-r)^{1/4})]\big)$, which extends to a homeomorphism of the closed square onto $\overline{\Omega_n}$ taking the edge $r=1$ onto I_n. A common point of $\overline{\Omega_n}$ and $\overline{\Omega_m}$, $n<m$, would give $\Delta([\alpha(n),\beta(n)],[\alpha(m),\beta(m)])\le 2\varepsilon_m^{1/4}$, against IV. $\operatorname{diam}\Omega_n\le\varepsilon_n+\pi\varepsilon_n^2+2\varepsilon_n^{1/4}$.

For $re^{i\theta}\notin\Omega_n$ either $r\le 1-\varepsilon_n$ or $(r>1-\varepsilon_n$ and $\Delta(\theta,[\alpha(n),\beta(n)])\ge(1-r)^{1/4})$. In the first case $P_r\le(1+r)/(1-r)\le 2/\varepsilon_n$ and $f_n\le\frac{1}{2\pi}\cdot\pi\varepsilon_n^2\cdot\frac{2}{\varepsilon_n}=\varepsilon_n$. In the second case, with $v=(1-r)^{1/4}<1$, for $t\in[\alpha(n),\beta(n)]$ we have $\cos(\theta-t)\le\cos v\le 1-\frac{v^2}{2}+\frac{v^4}{24}\le 1-\frac14\sqrt{1-r}$;

as $r > \frac{3}{4}$,

$$f_n \le \frac{1}{2\pi} \cdot \pi\varepsilon_n^2 \cdot \frac{1-r^2}{\frac{1}{2}r\sqrt{1-r}} \le \frac{8}{3}\varepsilon_n^2\sqrt{1-r} \le \varepsilon_n\sqrt{1-r} \le \varepsilon_n,$$

since $\varepsilon_n \le \frac{1}{4}$. This proves (1) and (2); (3) follows from (2) because $r \to 1$ along such an arc. □

Theorem 9.7 ([5]; answer to [9, Problem 5]). *There are a bounded harmonic function f in $\mathbb{D}$ and a boundary function $\varphi : C \to \mathbb{R}$ for f that is not of Baire class ≤ 1.*

Proof. With the notation of Lemma 9.6, let $\chi = \sum_n \chi_n$ (the indicator of $\bigcup_n I_n$) and $f = P[\chi] = \sum_n f_n$ (monotone convergence); f is bounded harmonic (Fact 9.1(i)).

χ is continuous with value 1 on each open arc I_n^*, so f has general limit 1 there (Fact 9.1(ii)) and, by Lemma 9.4, asymptotic value 1 at $e^{i\alpha(n)}$ and $e^{i\beta(n)}$.

Let $\zeta \in C \setminus \bigcup_n I_n^*$. By Lemma 9.3 there is an arc γ at ζ meeting no Ω_n. For $m \in \mathbb{N}$ and $z \in \gamma$, by (1) and I, $|f(z)| \le \sum_{n\le m} f_n(z) + \sum_{n>m} \varepsilon_n \le \sum_{n\le m} f_n(z) + 2^{-m-1}$, and by (3) $\limsup|f| \le 2^{-m-1}$ along γ. So f has asymptotic value 0 at ζ; in particular the endpoints of the I_n are ambiguous points. Let $A = \{e^{i\alpha(n)}\}$, $B = \{e^{i\beta(n)}\}$, and define $\varphi = 1$ on $\bigcup_n I_n^* \cup B$ and $\varphi = 0$ on the rest of C. Then φ is a boundary function for f.

By V, every p_h lies in $\bigcup_n I_n$, so $\bigcup_n I_n$ is dense in C. Given $e^{i\alpha(n)}$ and a short open arc O ending at it on the side away from I_n, some I_m meets O; since $I_m \cap I_n = \emptyset$, $e^{i\beta(m)} \in O$; and some $I_{m'}$ meets the open arc between $e^{i\beta(m)}$ and $e^{i\alpha(n)}$, so lies inside it and $e^{i\alpha(m')} \in O$. Arguing likewise at points of B, each point of $A \cup B$ is a limit of other points of A and of points of B. Hence $P = \mathrm{Cl}(A \cup B)$ is a nonempty closed (so complete) set in which both A and B are dense. As $\varphi = 0$ on A and $\varphi = 1$ on B, $\varphi|_P$ is continuous at no point. If φ were of Baire class ≤ 1, so would be $\varphi|_P$, contradicting Corollary 3.25. □

Theorem 9.8 ([5, Theorem]). *Let u be harmonic in $\mathbb{D}$ with $\{\int |u(re^{i\theta})|^p d\theta : 0 \le r < 1\}$ bounded for some $p > 1$. Let A be the set of points of C at which u has an asymptotic value and G the set at which u has a general limit. Then $A \setminus G$ is of the first category.*

Proof. Let S be the set where the radial limit ψ exists; $S \subseteq A$, $C \setminus S$ is null, and $u = P[\psi]$ (Fact 9.1(iii)). Extend $\psi|_S$ to a boundary function φ_1 on A. By Corollary 6.3 there is $\varphi \in \mathcal{B}_1(A, \mathbb{R})$ with $\varphi = \varphi_1$ off a countable set; then $\varphi = \psi$ almost everywhere and $u = P[\varphi]$. By Theorem 3.24 there is $K \subseteq A$ with $A \setminus K$ of the first category in A, hence in C (Lemma 3.26), such that $\varphi|_A$ is continuous at each point of K. By Fact 9.1(ii), $K \subseteq G$; so $A \setminus G \subseteq A \setminus K$. □

Corollary 9.9 ([5, Corollary]). *Under the hypothesis of Theorem 9.8, if φ is a boundary function for u defined on $E \subseteq C$, then the set of points of E at which φ is discontinuous is of the first category in C.*

Proof. At a point $\zeta \in E \cap G$ with general limit L: if $|u - L| < \varepsilon$ on $\mathbb{D} \cap S(\delta, \zeta)$, then every $x' \in E \cap S(\delta, \zeta)$ has $|\varphi(x') - L| \le \varepsilon$, since $\varphi(x')$ is a limit along an arc whose tail lies in $S(\delta, \zeta)$. So φ is continuous on $E \cap G$, and its discontinuities lie in $A \setminus G$. □

Corollary 9.10 (analytic functions; [5]). *Theorem 9.8 and Corollary 9.9 hold for holomorphic F in $\mathbb{D}$ with bounded p-means, $p > 1$.*

Proof. Apply them to $\mathrm{Re}\, F$ and $\mathrm{Im}\, F$, whose p-means are bounded by those of F: if F has an asymptotic value at x so do both parts, and if both have general limits so does F. □

Remark 9.11. The example of [9, Theorem 6] shows that the hypothesis on the integral means cannot be omitted.

10. A cube with a non-Borel set of curvilinear convergence

Source: [7]. For an open connected $\Omega \subseteq \mathbb{R}^n$ and $f : \Omega \to \mathbb{R}^m$, the set of curvilinear convergence $\mathrm{CC}(f)$ is the set of $p \in \partial\Omega$ for which there is an arc γ at p with f converging to a finite limit along γ.

Definition 10.1 (asymptotic values)**.** $a \in \mathbb{R}^m$ is an *asymptotic value* of $f : \Omega \to \mathbb{R}^m$ if there is a continuous $v : [0,1) \to \Omega$ with $\operatorname{dist}(v(t), \mathbb{R}^n \setminus \Omega) \to 0$ and $f(v(t)) \to a$ as $t \to 1$; it is a *point asymptotic value at p* if moreover $v(t) \to p \in \partial\Omega$. By Corollary 4.4, $p \in \mathrm{CC}(f)$ iff f has a (finite) point asymptotic value at p.

Fact 10.2 (analytic non-Borel sets)**.** *There is an analytic subset of $\mathbb{R}$ that is not a Borel set.*

Fact 10.3 (Kierst [15])**.** *For every analytic set $A \subseteq \mathbb{R}$ there is a holomorphic function h in $\mathbb{D}$ omitting the values i and $-i$ whose set of finite asymptotic values is $A \cup \{-i, i\}$.*

Fact 10.4 (Lehto–Virtanen [17, p. 53]; Bagemihl–Seidel [11, Theorem 1])**.** *If h is holomorphic in $\mathbb{D}$ and omits two finite values, then every finite asymptotic value of h is a point asymptotic value of h.*

Lemma 10.5 ([7, Lemma])**.** *There is a continuous $s : \mathbb{H} \to \mathbb{C}$ with $|s| < 1$ such that the set E of real asymptotic values of s lying in $(-1,1)$ equals the set of real point asymptotic values of s lying in $(-1,1)$, and E is not a Borel set.*

Proof. Let A be analytic and not Borel, h as in Fact 10.3, and $\rho(w) = w/(1+|w|)$, a homeomorphism of $\mathbb{C}$ onto $\mathbb{D}$ mapping $\mathbb{R}$ onto $(-1,1)$. Put $s(x,y) = \rho\big(h((1-y)e^{ix})\big)$ for $0 < y < 1$ and $s(x,y) = \rho(h(0))$ for $y \geq 1$. Let $b \in (-1,1)$. If b is an asymptotic value of s along $v = \langle x(t), y(t)\rangle$ (so $y(t) \to 0$), then $\rho^{-1}(b)$ is an asymptotic value of h along $(1-y(t))e^{ix(t)}$, hence lies in A. If $a \in A$, by Fact 10.4 a is a point asymptotic value of h at some e^{ip}, along a path $w(t) \to e^{ip}$; writing $w(t) = r(t)e^{i\vartheta(t)}$ with ϑ continuous (for t near 1), $\vartheta(t) \to p + 2\pi k$, and $\rho(a)$ is a point asymptotic value of s at $p + 2\pi k$ along $\langle \vartheta(t), 1 - r(t)\rangle$. Thus both sets equal $\rho(A)$, which is not Borel. □

Remark 10.6. [7] notes that the lemma can also be proved without meromorphic functions, from [21, Theorem 113], a theorem of Bagemihl–McMillan, and the methods of [6], but omits the "messy construction".

Theorem 10.7 ([7, Theorem])**.** *There is a bounded continuous $f : K^\circ \to \mathbb{C}$ on the open cube $K^\circ = (-1,1) \times (0,1) \times (-1,1)$ whose set of curvilinear convergence is not a Borel set. (So McMillan's theorem does not extend to three dimensions.)*

Proof. Let s, E be as in Lemma 10.5 and $g(x,y) = s(x/y, y)$ on Q.

E is the set of real point asymptotic values of g at $\langle 0,0\rangle$ in $(-1,1)$. If $g \to b$ along a path $\langle x(t), y(t)\rangle \to \langle 0,0\rangle$ in Q°, then $s \to b$ along $\langle x/y, y\rangle$, whose second coordinate tends to 0; so $b \in E$. If $s \to b$ along $\langle u(t), v(t)\rangle \to \langle p, 0\rangle$, then $g \to b$ along $\langle u(t)v(t), v(t)\rangle \to \langle 0,0\rangle$, which lies in Q° for t near 1.

Construction. Let $d_0(t)$ be the supremum of the $\delta \in (0,1]$ such that $z, z' \in Q$, $\operatorname{Im} z, \operatorname{Im} z' \geq t$, $|z - z'| < \delta$ imply $|g(z) - g(z')| \leq t$; let $d(t) = \min\{\frac{1}{4}d_0(t/2), \frac{1}{4}t\}$, and k as in Theorem 7.15 (so $|k| \leq \sqrt{2} \leq 2$). Here $d_0(t) > 0$ because g is uniformly continuous on the compact set $[-1,1]\times[t,1]$, and d_0 is weakly increasing (raising t shrinks the region and loosens the tolerance); so d is positive and weakly increasing, as Theorem 7.15 requires. Put $f(x,y,z) = (g(x,y) - z)\,k(x,y)$. For any three points $P = \langle x, y, z\rangle$, P', P'' of K°, writing $g' = g(x', y')$ etc.,

$$|f(P') - f(P'')| \geq |g - z||k' - k''| - 2|g' - g| - 2|g - g''| - 2|z - z'| - 2|z'' - z|, \tag{10.1}$$

because $f(P') - f(P'') = (g-z)(k'-k'') + \big((g'-z') - (g-z)\big)k' - \big((g''-z'') - (g-z)\big)k''$.

Let $L = \{\langle 0,0,z\rangle : -1 < z < 1\}$ and $F = \mathrm{CC}(f)$. We show $F \cap L = \{\langle 0,0,b\rangle : b \in E\}$; then F is not Borel, since otherwise $F \cap L$ would be.

If $b \in E$, take an arc γ at $\langle 0,0\rangle$ in Q° along which $g \to b$; along $\gamma' = \{\langle x,y,b\rangle : \langle x,y\rangle \in \gamma\}$, $g - z \to 0$, so $f \to 0$, and $\langle 0,0,b\rangle \in F$.

Conversely let f converge along an arc γ' at $\langle 0,0,b\rangle$ in K°, and let γ be its projection to the $\langle x,y\rangle$-plane, a path ending at $\langle 0,0\rangle$ and otherwise in Q°. We claim $g(x,y) - z \to 0$ along γ'; then $g \to b$ along γ, so b is a point asymptotic value of g at $\langle 0,0\rangle$ and $b \in E$. If not, there are $\varepsilon > 0$ and points $\langle x_n, y_n, z_n\rangle \in \gamma'$ converging to $\langle 0,0,b\rangle$ with $|g(x_n,y_n) - z_n| \geq \varepsilon$. Since the only point of γ' with $y = 0$ is its endpoint, there is $\delta > 0$ such that points of γ' with $y \leq \delta$ have z-coordinates within $\varepsilon/8$ of each other. Fix n large, with $y_n < \min\{3\varepsilon/32, 3\delta/4, 3/4\}$ and with the projection of the other endpoint of γ' farther than y_n from $\langle x_n, y_n\rangle$, and put $a = 4y_n/3$. By Lemma 4.5(b) applied to (a parametrization of) γ, there is a connected $\gamma^* \subseteq \gamma \cap \overline{S(d(a), \langle x_n, y_n\rangle)}$ containing $\langle x_n, y_n\rangle$ and a point at distance $d(a)$ from it. As $d(a) \leq a/4$, $\gamma^* \subseteq [-1,1] \times [a/2, a]$, and Theorem 7.15 gives points $\langle x_n', y_n'\rangle$, $\langle x_n'', y_n''\rangle$ of γ^* with $|k' - k''| \geq 2$; lift them to points of γ' with third coordinates z_n', z_n''. These have $y \leq a < \delta$, so $|z_n' - z_n|, |z_n'' - z_n| < \varepsilon/8$; and $|g(x_n,y_n) - g'|, |g(x_n,y_n) - g''| \leq a/2 < \varepsilon/16$ by the choice of d. By (10.1), $|f(P_n') - f(P_n'')| \geq 2\varepsilon - \frac{\varepsilon}{4} - \frac{\varepsilon}{2} > \varepsilon$. Since $P_n', P_n'' \to \langle 0,0,b\rangle$ along γ', f does not converge along γ': a contradiction. □

Remark 10.8. [7] leaves open whether a continuous *real*-valued function in a cube can have a non-Borel set of curvilinear convergence.

Appendix A. The dissertation's special families

The dissertation uses a more restrictive notion of special family than [6]. Its results are proved here for completeness; they are not used elsewhere. A *complementary interval* of a closed $F \subseteq \mathbb{R}$ is a component of $\mathbb{R} \setminus F$.

Definition A.1 ([3, §4])**.** A *thesis-special family* is a nonempty family $\mathcal{F}$ of bounded closed subsets of $\mathbb{R}$ with an enumeration $F_1, F_2, \ldots$ (every member occurring) such that for $m > n$, either F_m lies in a complementary interval of F_n or F_m splits with F_n.

Lemma A.2. *Any two members of a thesis-special family are equal, disjoint, or split. Hence Lemmas 7.8 and 7.9 apply to thesis-special families of subsets of X (*[3, Lemmas 13, 14]*).*

Proof. A set lying in a complementary interval of F_n is disjoint from F_n. □

Lemma A.3 ([3, Lemma 9])**.** *Every F_σ set $E \subseteq \mathbb{R}$ is $\bigcup_n E_n$ for bounded closed E_n any two of which are disjoint or split.*

Proof. As in Lemma 7.5, with $\mathbb{R}$ in place of X; enumerate the resulting countable family, padding with $\emptyset$ if it is finite (repeating a member with two points would violate the condition). □

Lemma A.4 ([3, Lemma 10])**.** *Every F_σ set $E \subseteq \mathbb{R}$ is the union of a thesis-special family.*

Proof. Take E_n from Lemma A.3. We construct $1 = n_1 < n_2 < \cdots$ and bounded closed sets $F_1, F_2, \ldots$ such that for every s: (a) if $n_s \geq r > t \geq 1$, then F_r lies in a complementary interval of F_t or splits with F_t; (b) each F_i, $i \leq n_s$, lies in some E_j, $j \leq s$; (c) $\bigcup_{i \leq n_s} F_i = \bigcup_{j \leq s} E_j$. Start with $F_1 = E_1$. Given stage s, let $\mathcal{J}$ be the set of complementary intervals of the bounded closed set $\bigcup_{j \leq s} E_j$.

E_{s+1} meets only finitely many $I \in \mathcal{J}$. Otherwise pick $x_m \in I_m \cap E_{s+1}$ for infinitely many distinct I_m; a subsequence is strictly monotone, say increasing, with limit x. Between consecutive terms lies an endpoint a_k of some I_m, which belongs to $\bigcup_{j \leq s} E_j$; infinitely many a_k lie in one

E_u, so $x \in E_u \cap E_{s+1}$. Then E_u and E_{s+1} split at x; but both contain points less than x, which is impossible.

Let the new sets $F_{n_s+1}, \dots, F_{n_{s+1}}$ be $\emptyset$ and the sets $\overline{I} \cap E_{s+1}$ for the finitely many $I \in \mathcal{J}$ meeting E_{s+1}. (We use the closure $\overline{I}$, so that the new sets are closed; with I itself, as written in [3], they need not be.) (b) and (c) are clear. For (a) with r new: if t is old, $F_t \subseteq E_v$ with $v \le s$. If $E_v \cap E_{s+1} = \emptyset$, let J be the smallest closed interval containing $F_r = \overline{I} \cap E_{s+1}$; $J^* \subseteq I$ misses E_v, and the endpoints of J lie in E_{s+1}, so $J \cap E_v = \emptyset$ and F_r lies in a complementary interval of F_t. If E_v and E_{s+1} split, so do F_t and F_r. If t is new too, F_r and F_t come from distinct $I_1, I_2 \in \mathcal{J}$, and $\overline{I_1}, \overline{I_2}$ split. Finally $\mathcal{F} = \{F_k\}$ is thesis-special with $\bigcup \mathcal{F} = E$. □

Lemma A.5 ([3, Lemmas 11, 12])**.** *If $\mathcal{F}$ and $\mathcal{G}$ are thesis-special, so is $\mathcal{F} \wedge \mathcal{G} = \{F \cap G : F \in \mathcal{F}, G \in \mathcal{G}\}$, and $\bigcup(\mathcal{F} \wedge \mathcal{G}) = \bigcup \mathcal{F} \cap \bigcup \mathcal{G}$.*

Proof. Enumerate $\mathbb{N} \times \mathbb{N}$ as $(a(k), b(k))$ by increasing $a + b$ and, for equal sums, increasing b; then $k < l$ implies $a(k) < a(l)$ or $b(k) < b(l)$. Put $H_k = F_{a(k)} \cap G_{b(k)}$. If $l > k$ and $a(k) < a(l)$, then $F_{a(l)}$ lies in a complementary interval of $F_{a(k)}$, which lies in a complementary interval of $H_k \subseteq F_{a(k)}$, or $F_{a(l)}$ splits with $F_{a(k)}$; in either case H_l relates to H_k in the same way. The case $b(k) < b(l)$ is symmetric. □

Appendix B. Concordance with the sources

Source item	Here
[4], with [8]	
Sutcliffe Thm 2	Prop. 2.3, Prop. 2.4, Thm. 2.6
Sutcliffe Thm 3	Thm. 2.7
Lemma 1, Lemma 2	Lemma 2.8, Lemma 2.9
Theorem, Corollary	Thm. 2.10, Cor. 2.11
[1] (from the review)	
Theorem 1 (homeomorphisms)	Thm. 6.7 (new proof)
Theorems 2, 3	Cor. 6.3
Theorems 4, 5	Thm. 8.3, Cor. 8.4
$f = 0$ a.e. with given boundary function	Cor. 8.12
remarks on three dimensions	not a theorem; see Section 10
[2]	
Lemma 1	Lemma 3.34
Lemmas 2, 3	Lemmas 5.2, 5.3
Theorem 1 (McMillan)	Thm. 5.4
Theorem 2, Corollaries 1–4	Thm. 5.5, Cor. 5.6
remark (E of type G_δ)	Prop. 6.5
[3]	
Lemmas 1–6	Lemma 3.4, 3.10, 3.11, 3.12, 3.13, Thm. 3.14
Theorems 1, 2	Thm. 3.20, Thm. 3.22
Lemmas 7, 8; Theorem 3	Lemmas 5.8, 5.9; Thm. 5.7
Lemmas 9–12	Lemmas A.3, A.4, A.5
facts (6)–(17); Lemmas 13, 14	Lemma 7.3; Lemmas 7.8, 7.9, A.2

Theorem 4	Thm. 7.11 (and Thm. 7.10)
Lemma 15; Theorem 5 and Corollary	Lemma 6.1; Thm. 6.2, Cor. 6.3
Lemmas 16, 17; Theorems 6, 7	Lemma 7.13, 7.12; Thm. 7.14, 7.19
Lemmas 18–22	Lemmas 3.33, 3.34, 3.36, 3.37, 3.38
Theorem 8 and Corollary	Thm. 8.3, Cor. 8.4
Lemmas 23, 24; Theorem 9; example	Lemmas 8.6, 8.7; Thm. 8.8; Prop. 8.9
Lemma 25; Theorem 10, Corollaries	Lemma 8.10; Thm. 8.11; Cor. 8.12, 8.14
unsolved problems 1–5	open questions; not formalized
[5]	
Lemma	Lemma 9.3
example (answer to [9, Problem 5])	Lemmas 9.5, 9.6, Thm. 9.7
Theorem, Corollary, analytic case	Thm. 9.8, Cor. 9.9, Cor. 9.10
[6]	
(A) with (I)–(XXII)	Thm. 7.10; Def. 7.4, Lemmas 7.5, 7.6; Def. 7.1, Lemma 7.3; Def. 7.7, Lemmas 7.8, 7.9
(B) with (I)–(IV)	Thm. 7.14; Lemmas 7.12, 7.13; Thm. 3.14
(C), (D), (E), (F)	Thms. 7.15, 7.16, 7.17, 7.18; Def. 3.17, Lemma 3.18
Appendix (a), (b), (c)	Thms. 3.20, 3.22, 3.21
Problems 1, 2	open questions; not formalized
[7]	
Lemma, Theorem	Lemma 10.5, Thm. 10.7
Problem	open question; not formalized

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